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Mass Reproducibility in Psychological Science

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Mass Reproducibility in Psychological Science

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September 2026

INSTITUTE for REPLICATION

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Mass Reproducibility in Psychological Science

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Mass Reproducibility in Psychological Science

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Abstract

Concerns about the credibility and reproducibility of published findings have motivated numerous initiatives to improve transparency in psychology. In this article, we present evidence from a large-scale reproducibility initiative conducted in partnership between *Psychological Science* and the Institute for Replication (I4R). Our project systematically reproduces empirical results published in the journal between 2024 and 2025 using independent research teams. Through a combination of Replication Games and researcher-led initiatives, 67 independent teams examined 44% of the articles (*i.e.*, 64 articles) published during our time frame, documenting data and code availability, checking for coding errors, and deviations from preregistration, and computationally reproducing the main numerical results. We find that the majority of articles provide cleaned data and analysis scripts, though raw data and data-cleaning scripts are shared far less often. Computational reproducibility depends strongly on the starting point: 58.2% of articles are fully reproducible from final analysis data, but only 28.4% are fully reproducible from raw data. Teams identify coding errors or statistical reporting inconsistencies in 63% of articles, although most issues are minor; 40.5% of the articles flagged with coding or statistical reporting issues contain discrepancies with potentially major implications for estimates or interpretation. Preregistration is common (74.6% of articles), yet 84% of preregistered articles deviate from their pre-analysis plan, and nearly half of these deviations go unacknowledged. Our findings provide new insights into the reliability of empirical results in a leading psychology journal and highlight the value of structured mass reproducibility initiatives.

Keywords: reproducibility, replication, open science, psychology, research transparency

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1 Research Transparency Statement

General Disclosures

Independence and relationship with Psychological Science: Abel Brodeur, the study's PI, exchanged emails and met once by Zoom with Simone Vazire in the months before she became Editor-in-Chief of Psychological Science. These discussions were unrelated to the design or implementation of the present study. Neither Brodeur nor other members of the research team subsequently interacted with Vazire or anyone else at Psychological Science regarding this project until summer 2026, when the authors shared a draft of the manuscript with Vazire and the STAR team. They did not provide comments or other feedback on the manuscript and instead invited the authors to submit the paper to Psychological Science. The authors did so in early September 2026. The project was conceived, designed, and conducted independently of Psychological Science. The journal provided no financial support or other resources for the project and had no role in selecting the articles included in the study, defining the reproduction protocol, conducting or evaluating the reproductions, analyzing the results, or interpreting the findings. The journal did not provide the research team with a list of eligible studies, replication packages, data, code, or other research materials. All decisions regarding study selection, methods, analysis, and reporting were made independently by the research team. The Institute for Replication has subsequently collaborated with Psychological Science on a separate follow-up reproducibility initiative. This collaboration began independently of the design and execution of the study reported here and did not affect the study sample, procedures, analyses, or conclusions. We disclose both this subsequent collaboration and our planned submission of the manuscript to Psychological Science because the journal itself is the subject of the present evaluation and therefore has an institutional interest in the findings.

Conflict of interest: Abel Brodeur: Chair of the Institute for Replication. The Institute for Replication's core mission is to promote data sharing and computational reproducibility. Amélie Gourdon-Kanhukamwe: Based on Acem et al. (2025), my career development may benefit from publishing the study; I do not have other professional roles that stand to benefit from the current study. Daryl Y. H. Lee: Serve as a reproducibility checker for the REPEAT Network, which conducts computational reproducibility checks for Psychological Science; none of the articles examined in this meta paper were among those he checked. I declare no other conflicts of interest. Lukas Röseler: I declare that I was one of the organizers of the Replication Games event where I was part of a team that carried out a reproduction study. Mahmoud Elsherif: We hereby declare that we have no interests that may come into conflict with publishing or promoting the current study. The study does not have any direct or indirect financial implications for the authors, beside that the work has been carried out as the authors' professional activities at their indicated affiliations. The current work continues previous published research on a similar topic. Based on Acem et al. (2025), our research career development and esteem may benefit from publishing the study. The authors do not have other professional roles that stand to benefit from the current study.

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ported by their grants also generated important spillovers for this project. In particular, we were able to integrate interested psychology reproducibility teams into Replication Games funded by Coefficient Giving organized for our non-psychology projects.

Artificial intelligence: Abel Brodeur: Artificial intelligence assistance was used for code generation and checking, and for proofreading this manuscript. Furthermore, artificial intelligence assistance was used for some of the 67 reproducibility reports. Aleksander Psurek: Programming (code), research on statistical models, search for related literature. Aleksandra Pytlińska: Translation and review, proofreading. Beata E. Andrzejewska: Translation and review, assistance, proofreading. Benedikt Stöckle: Coding assistance. Carolin Mundinger: Text editing / improving writing style. Charly Marie: Text editing / improving writing style (DeepL). Christoph Schild: Code development, formulation and refinement of text passages, and proofreading (Generative AI). Daniel Woods: transcribing R code that was provided as a pdf in the replication package to actual R code - accuracy confirmed by replicated output. Daryl Y. H. Lee: Reproduction analyses, drafting, and internal review of the report (Claude Code and Codex); the authors verified all analyses and results and take full responsibility for the content. Richard D. Brown: Proofreading report (ChatGPT). Ensar Acem: Proofreading. Federico D'Atri: Proofreading and coding assistance. Harry D. Coulson: Proofreading and coding assistance. Hemu Xu: Reproduction analyses, drafting, and internal review of the report; the authors verified all analyses and results and take full responsibility for the content (Claude Code and Codex). Jason Collins: Original paper code review, coding assistance, team report editing and proof reading. Jia-Shun Wang: Coding assistance and light copyediting. Joanna M. Rutkowska: Proofreading and coding assistance. Joris Frese: Coding assistance and light copyediting. Joséphine C.P. Levy: Coding assistance. Joshua B. Miller: Original paper code review, coding assistance and light team report copyediting. Karlo Doroc: Original paper code review, coding assistance and light team report copyediting. Linda Arrighi: Coding assistance. Lirola Rose-Marie: Translation. Lukas Robert Braun: Coding assistance. Lukas Röseler: Coding assistance. Max R. P. Grossmann: Copy-editing and coding assistance. Melissa Bretthauer: Support code development, formulation and refinement of text passages, and proofreading (Generative AI). Michael Keinprecht: Coding assistance including explanation of original code. Mirela Zaneva: Double-checking code. Nicole C. Engeler: Proofreading and coding assistance. Ole Hätscher: Support code development, formulation and refinement of text passages, and proofreading (Generative AI). Pablo Rodrigo Grassi: Coding assistance. Peter Szecsi: Proofreading and coding assistance. Roy Karl Eberle: Coding assistance. Ruben Laukenmann: Coding assistance. Sharon Sze Long Chung: Reproduction analyses, drafting, and internal review of the report; the authors verified all analyses and results and take full responsibility for the content (Claude Code and Codex). Shoucong Carol Xiong: Copy-editing and coding assistance. Simon M. Breil: Support code development, formulation and refinement of text passages, and proofreading. Yuhe Jin: Coding assistance. Yuri G. Pavlov: Copy-editing and coding assistance. David Valenta: Coding assistance.

2 Introduction

The credibility of empirical research depends on the ability of independent researchers to verify published findings. Concerns about whether a substantial share of published findings cannot be independently verified have motivated the credibility revolution in psychological science (Nosek et al., 2018; Open Science Collaboration, 2015; Simonsohn et al., 2014; Vazire, 2018). These concerns have made the verifiability of published research an object of study in its own right, raising the question of whether, and under what conditions, empirical findings can be independently confirmed.

Verification takes more than one form. In the scientific literature, a distinction is commonly made between *computational reproducibility* and *replication*. Computational reproducibility refers to the ability to obtain the same results reported in a study when using the original data and analytical procedures. Replication, in contrast, involves collecting new data and testing whether the original findings hold in an independent sample using similar research designs (Dreber and Johannesson, 2025). Beyond whether results can be reproduced at all, a related concern is whether the reported results are correct: coding errors and reporting inconsistencies can affect the validity of published numbers even when the analysis is otherwise reproducible. A further dimension concerns the transparency of the research process itself: whether the analyses reported correspond to those the original authors specified in advance in a preregistration. These properties each bear on the credibility of a finding, yet they are conceptually distinct and need not coincide.

In this article, we report results from a mass reproducibility effort conducted through a partnership between *Psychological Science* and the Institute for Replication (I4R). The goal of this initiative is to systematically reproduce empirical findings published in the journal and document the prevalence of computational reproducibility, coding and reporting issues, and deviations from preregistered analysis plans (Hardwicke and Vazire, 2024). We classify the coding errors and reporting inconsistencies teams uncover into distinct types—spanning data-processing, implementation, model-specification, and sample-construction errors as well as documentation and reporting problems—and we sort deviations from preregistration by the aspect of the analysis they affect, covering estimation and modeling, variable construction, sample size and data collection, data exclusions, and design or reporting scope, and we report whether each deviation was disclosed in the published article.

In general, verifying these properties at scale has only recently become feasible. Over the past decade, journals, professional societies, and research communities have implemented reforms aimed at improving transparency, data sharing, and methodological rigor, including preregistration, open-data requirements, and transparency standards across journals (Chambers, 2013; Hardwicke et al., 2022; Kidwell et al., 2016; Munafò et al., 2017; Nosek et al., 2015, 2018). These reforms contribute to generating the preregistrations, data, and codes that independent researchers need to assess both whether reported results can be reproduced and whether the analyses behind them match what was planned or has coding flaws. A growing body of work has taken up this opportunity: large-scale replication efforts have examined whether published findings hold in new samples (Camerer et al., 2016, 2018; Klein et al., 2014; Open Science Collaboration, 2015; Tyner et al., 2026),

while complementary initiatives have assessed computational reproducibility by verifying whether published results can be regenerated from the original data and code (Brodeur et al., 2026b; Chang and Li, 2022; Fišar et al., 2024; Gertler et al., 2018; Hardwicke et al., 2022; Miske et al., 2026). Together, these projects provide systematic evidence on the reliability and transparency of published research and help identify barriers to reproducibility such as incomplete documentation, unavailable data, or coding errors.

A parallel literature has established that deviations from preregistered analysis plans are common and frequently undisclosed. In an early cohort of preregistered studies in *Psychological Science*, 24 of 27 contained at least one undisclosed deviation (Claesen et al., 2021), and later work across psychology has found that preregistrations often lack the detail needed to constrain analysis and that undisclosed deviations remain frequent (van den Akker et al., 2024). Similar rates appear in adjacent fields, where most deviations are not linked to questionable research practices (Torka and Hüffmeier, 2025), and many preregistrations omit core procedural elements, such as planned analyses, sample size, or exclusion criteria, making it difficult to determine whether the reported analysis deviates from what was originally planned (Hahn et al., 2025).

Our contribution is threefold. First, we assess computational reproducibility and adherence to preregistration through a structured, mass reproducibility audit, in which independent teams evaluate each article against its replication package and preregistration using a standardized protocol, with original authors given a chance to respond—an approach that grounds each rate we report in a direct, per-article verification.[1] Second, we examine multiple dimensions of research transparency jointly for the same set of articles: we distinguish reproducibility from two starting points, raw and final analysis data; document coding and reporting errors; and evaluate adherence to preregistration. Typically, these dimensions are examined separately rather than together on a single sample. Third, we characterize the reproducers themselves—their backgrounds, experience, and attitudes toward open science—both to document who performs this verification work and how they differ from the original authors, and to examine whether their prior attitudes are associated with the outcomes they report. Because our teams self-selected, we treat the reproducer pool as an object of study, while recognizing that studying a volunteer sample can help us characterize this selection bias but not remove it.

This article presents results from the first wave of reproductions conducted under this partnership, focusing on articles published between 2024 and 2025. None of our analyses were preregistered, and all findings should therefore be interpreted as descriptive and exploratory rather than confirmatory. We use two complementary streams to generate reproduction reports: structured Replication Games, in which researcher teams collaborate during organized workshops, and researcher-led initiatives, in which independent teams self-select articles to reproduce asynchronously. Across both streams, 67 independent teams documented data and code availability, checked coding errors and deviations from preregistration, and computationally reproduced the

[1] A replication package refers to the set of materials an author deposits to allow others to reproduce the results reported in an article, typically comprising the data (raw and/or analysis-ready), the code used to clean the data and generate the reported results, and documentation such as a README file describing how the materials fit together.

main numerical results.

The interpretation of our findings should take into account the timing of the studies included in our sample. Approximately 84.4% of the reproduced articles were submitted before the arrival of the current editorial board at *Psychological Science* and were, therefore, not subject to the journal's recent transparency and reproducibility initiatives, including the stronger expectations regarding data, code, and materials sharing articulated by Hardwicke and Vazire (2024). As a result, the reproducibility outcomes documented in this article partly reflect practices that predate these reforms. Because our design is a single cross-section rather than a before-and-after comparison, we do not attempt to attribute reproducibility outcomes to these policies. The estimates reported here should be viewed as a baseline against which future waves of reproductions, covering articles submitted under the new reform, can be compared to assess whether those expectations translate into measurable change, rather than a definitive assessment of the impact of the journal's current policies.

3 Methods

This project was designed to systematically evaluate the computational reproducibility of empirical findings published in *Psychological Science* and their adherence to preregistered analysis plans. The primary goal was to generate a large number of independent reproduction reports conducted by external, volunteer research teams using transparent and standardized procedures.

To encourage broad participation while maintaining high methodological standards, I4R invited graduate students and researchers holding a PhD to participate in the project. On average, research teams had 3.358 reproducers (members). Research teams that successfully completed a reproduction report were offered coauthorship on this meta-paper summarizing the collective findings. This model incentivizes careful and transparent reproduction work while enabling the aggregation of evidence across multiple independent teams (Brodeur et al., 2023).

Eligible studies consisted of all research articles published in *Psychological Science* between 2024 and 2025. In total, 157 items were published in the journal during the specified study period. Of these, 146 were full research articles and were therefore eligible for assignment to reproduction teams. The remaining 11 publications consisted of research notes, editorials, or errata and were excluded from the mass reproduction initiative. Consequently, our analyses focused on the 146 eligible research articles. The set of reproduced articles was not selected on the basis of research topic, findings, or methodological characteristics. Rather, assignment to our sample of eligible articles primarily depended on the minimum availability of materials required to conduct a computational reproduction. As a result, the completed reports should be interpreted as reflecting the subset of articles for which independent verification is feasible using publicly available resources.

This is an ongoing project and many articles are still being scrutinized. At the time of writing, independent teams completed reproduction reports for 64 of the 146 eligible articles published in 2024–2025. Broken down by year, 55 of 94 eligible articles published in 2024 had completed reports. The remaining 39 articles have not yet been reproduced; of these, 19 lacked publicly available data, code, or both, rendering computational reproduction infeasible without direct contact with

the original authors or access to proprietary datasets. Our findings, especially those related to computational reproducibility, should thus be seen as portraying patterns among the majority of articles with data and code, but excluding about 20% of articles with no data, code or both.

Work on articles published in 2025 remains ongoing. A total of 52 eligible articles were published during that year, of which 9 had completed reproduction reports. This partial coverage of 2025 does not undermine our aims. Because we characterize current reproducibility, coding errors, and preregistration practice rather than estimate the effect of the journal's policy change, we do not require the complete set of post-reform articles that a before-and-after evaluation would demand. The 2025 articles reproduced so far contribute to this descriptive picture, and fuller coverage, together with the pre-reform baseline established here, will enable that policy evaluation in future waves. Figure 1 summarizes the flow of eligible articles through the reproduction process. A publicly available list of eligible papers was maintained on the Open Science Framework at https://osf.io/htc5b/. Each participating team selected an article from this list, or in some cases was assigned one by the I4R team (for example, to match an article's software requirements with a team's expertise), and registered its reproduction effort with I4R. The full list of articles reproduced is presented in Appendix B.

Teams were asked to follow a standardized protocol and adhere to a template (https://osf.io/8dkxc/). They first documented the completeness of the data and code(s), and then attempted to computationally reproduce the main claims reported in the article. During their work, they also documented any discrepancies, computational issues, coding errors or deviations from preregistered analysis plans. Teams were also asked, where feasible, to conduct robustness checks: re-analyses that deliberately depart from the original specification to test whether the reported findings hold under reasonable alternative analytical choices (*e.g.*, altering control variables, estimation methods, or sample restrictions). The majority, 52 teams, did so. Unlike the computational reproduction, which targets a common, well-defined outcome, robustness checks were tailored by each team to the design of its assigned article, so both the specific alternative tested and the criterion for "robustness" varied from paper to paper. We do not summarize them in this article, but they are available in each team's reproduction report.

Articles published in *Psychological Science* often report multiple studies. Due to time constraints, teams had the option to focus on only a subset of the studies in the reproduced article. In practice, 60 (89.552%) teams reproduced all the studies, while the remaining 7 teams completed an average of 4.143 studies per article, out of an average of 6.286 studies in the original articles assigned to these teams.

All reproduction reports were reviewed by I4R before inclusion in the final dataset. The reports were then shared with the original authors, who were invited to provide comments, clarifications, and feedback. Teams were given the opportunity to revise their reports in light of this feedback, and original authors were subsequently allowed to update their responses if revisions were made. All reproduction reports and author responses are publicly available (see Appendix B).

3.1 Generating Reproduction Reports

We used two complementary streams to generate the 67 reproduction reports for 64 articles published in *Psychological Science*. First, reports were produced through Replication Games organized by I4R. These structured reproducibility hackathons bring together researchers (*i.e.*, reproducers) who collaborate to reproduce published studies using supplementary materials, including preregistrations, data and scripts for the reproduction of the article. During these events, reproducers work in teams to independently verify the results reported in selected articles and document the steps required to reproduce the analyses. Teams are expected to reproduce all studies within an article where possible, and to prioritize a subset when time does not permit. Second, we invited independent research teams through social media posts to select an article and submit a reproduction report. The initiative was also promoted in Hardwicke and Vazire (2024).

The discrepancy between the 67 reports and 64 articles arises because 3 articles were each independently reproduced by two teams. We treat each of these reports as a separate unit of observation as this choice reflects how the reproductions were conducted: because teams were free to focus on a subset of an article's studies when time was limited, two teams assigned the same article may examine different studies or different results within it. For these articles, the two reports are better understood as complementary. Since only 3 of the 64 articles were independently reproduced by two teams, the treatment of these articles has little bearing on the aggregate patterns we report.

Article selection was guided by the feasibility of reproduction and familiarity with the programming language rather than by (our) expected results, which were not known when teams chose their article. Moreover, once a team selected an article it was committed to completing the reproduction regardless of difficulty or whether result failed to reproduce. Teams also had little incentive to abandon an article on the basis of its results: our protocol explicitly accommodates partial or failed reproductions, and inclusion of an article's reproduction report in this meta-paper did not depend on whether the assigned article's results were successfully reproduced. This limits the possibility that the completed set was filtered by teams abandoning articles whose findings did not reproduce. We nonetheless cannot rule out that teams favored more tractable articles at the selection stage.

3.1.1 Replication Games

A central component of our project is the use of Replication Games organized by I4R (Brodeur et al., 2023). Replication Games are structured workshops designed to systematically evaluate the computational reproducibility, robustness, and even replicability of published articles.

Replication Games typically involve between 25 to 125 researchers working in small teams of three to five participants.[2] For the present project, participants were first assigned to teams and provided a list of eligible articles in advance. Each team then selected an article and prepared for the event by reviewing the article and its replication materials. Teams developed a plan outlining the steps needed to reproduce the analysis and divide responsibilities among team members (see Appendix A). On the day of the event, participants, gathered in person or joining online, worked

[2]See I4R's website for a list of Replication Games (https://www.i4replication.org/).

intensively to reanalyze the article's data and verify the reported findings. For many empirical papers, much of the reproduction work can be completed during this workshop format. However, the event itself does not mark the end of the process: teams could continue working for several weeks or, when necessary, several months afterward to complete additional analyses, resolve outstanding issues and revise their report in response to feedback from the original authors.

Teams first attempt to computationally reproduce the main results and then turn to documenting deviations from the preregistration and conducting re-analyses. Reproductions focus on computational reproducibility and involve re-running the original analyses using the same data and code to determine whether the reported results can be obtained (*i.e.*, whether the reproduced estimates match in direction, magnitude, and statistical significance). Re-analyses, in contrast, assess robustness by modifying aspects of the original analysis to test whether the results hold under reasonable variations in research design. For example, teams might modify control variables, or use alternative datasets to evaluate the robustness of the original findings (Brodeur et al., 2023). We do not present the results of the re-analyses in this article.

After the workshop, teams complete a templated reproduction report describing their procedures, findings, and any discrepancies identified during the process, including missing documentation, computational difficulties, or coding errors. In total, 57 reports were completed using this stream.

3.1.2 Independent Reproduction Teams

In addition to Replication Games, we invited independent researchers to submit reproduction reports. Interested teams contacted I4R and selected an article from *Psychological Science* to reproduce. In total, 10 reports were completed using this stream.

3.2 Communication With Original Authors

Communication with original authors is an important component of the reproduction process. When reproduction teams encountered missing materials, unclear procedures, or discrepancies between reported results and reproduced outputs, I4R's chair (Abel Brodeur) contacted the authors of the original article.

In most cases, teams first attempted to reproduce results using publicly available materials, including data, analysis code, and supplementary documentation. When materials were incomplete or ambiguous, I4R sent a standardized email describing the reproduction effort and requesting clarification or additional materials.

Correspondence with original authors typically addressed one or more of the following issues: (1) requests for access to data or analysis code that were not publicly available; (2) clarification regarding data construction or variable definitions; (3) questions about sample restrictions or preprocessing steps; and (4) assistance identifying the exact code or procedures used to generate specific tables or figures.

When authors provided additional materials or clarifications, reproduction teams incorporated this information into the reproduction process and documented the changes in their reproduction reports. After the reports from both streams were finalized, they are then shared with the original authors for comments before being released publicly and later included as part of this meta-paper (Brodeur et al., 2023).

3.3 Dataset

Our final dataset consists of 67 reproduction reports covering 64 articles, with 3 articles independently reproduced by two teams each. In addition to the reports, we administered a survey to all participating reproducers collecting information on their background, research experience, and attitudes toward open science practices (see Appendix A). This survey also collected article-level information, including the programming language used in the original replication package, the runtime of the original code, and the quality of the replication packages. We further assembled information on the characteristics of the articles and their original authors (*e.g.*, number of authors and their Google Scholar citations).[3]

3.3.1 Original Authors and Reproducers

Table 1 summarizes the characteristics of original authors and reproducers. Original authors tend to hold more senior academic positions than reproducers. Approximately 38.7% of original authors are associate or full professors, compared with 9.7% of reproducers. In contrast, graduate students constitute the largest share of reproducers, accounting for 50.7% of the sample. These differences are also reflected in research visibility and experience. Original authors have substantially higher Google Scholar citation counts, with an average of 6,619 citations (median = 774.5) compared with 1,363 citations (median = 104) among reproducers. Most reproducers (71.1%) report a psychology-related field of research, while the remainder come from a range of disciplines such as economics, political science, sociology, marketing, and communication science.

The survey responses further provide information on reproducers' research practices and attitudes toward open science. Only 38.7% report having previously produced a replication package. On average, reproducers report having preregistered 7.8 studies (median = 3), although there is substantial variation across them ($SD = 14.8$). Reproducers' engagement also extended well beyond the replication games or any single workshop session: on average, they report spending 5.8 active days (SD = 4.7) on their reproduction report, indicating that these reports reflect sustained effort rather than a single rushed session.

3.3.2 Reproduced Articles

Table 2 summarizes the characteristics of the reproduced articles and their replication materials. On average, articles include 4.2 authors and 3.4 studies, while reproduction teams consist of 3.4 repro-

[3]When Google Scholar citations were unavailable, we used an alternative source such as ResearchGate.

ducers on average. R is the dominant programming language used in both the original articles and reproduction reports. Approximately 83.6% of original authors use R in their replication packages, compared with 89.6% of reproduction teams.

The computational burden associated with reproducing the analyses is generally modest. On average, the original code requires 2.3 hours to run from initial data preparation to the completion of all reported results (excluding time spent fixing the code to make it run), although computational times vary substantially across articles. As shown in Figure 2, this variation is driven by a small number of packages: runtimes are well under an hour for most articles but rise steeply at the upper tail, with two outliers requiring roughly 22 and 40 hours. Excluding these two outliers, the average runtime falls to 1.4 hours. Documentation quality varied across replication packages. Reproducers rated each package's README based on their overall judgment of whether it provided enough guidance to navigate and run the package: a clear README accompanied 23.9% of packages, 43.3% included a README that reproducers judged unclear, and 32.8% included no README. Because these ratings are subjective, they should be read as a proxy for documentation quality. Overall, these figures suggest that inadequate documentation remains a common obstacle to efficient computational reproduction.

3.3.3 Measures and Classifications

For each reproduced article, we draw on two kinds of measures: those recorded directly by the reproduction teams following our standardized protocol, and those we constructed afterward based on the information provided by the teams. We describe each measure below.

Data and Code Availability: Reproducers documented the availability of four types of materials: *Raw data* refers to the original, unprocessed dataset; *cleaning scripts* refer to codes used to transform the raw data into the dataset used for analysis; *final or analysis data* refers to the processed dataset used directly in the empirical analysis; and *analysis scripts* refer to the code used to generate the statistical results reported in the article, including tables and figures. Because articles often contain multiple studies drawing on different datasets, data and code availability may be complete for all studies, only a subset of studies or none of the studies. Reproducers were, thus, asked to classify each material type as *fully available* when they are provided for all studies within an article, *partially available* when they are provided for only some studies or are incomplete (*e.g.*, missing code components or data for certain variables), and *unavailable* when they are not provided for any study. This approach allows us to capture variation in transparency practices at the article level and avoid issues related to the number of studies varying across articles.

Computational Reproducibility: Computational reproducibility refers to the ability to obtain the same numerical results reported in the published article when using the same data and analytical procedures (code). Negligible differences attributable to rounding, software versions, or computational precision were not treated as reproduction failures. Reproduction teams attempted to com-

putationally reproduce the main empirical results, including the key statistical estimates presented in tables and figures as well as summary statistics when relevant. For each reproduction report, teams first attempted to run the original analysis code using the available data and documentation. When code was unavailable or incomplete, some teams attempted to reconstruct the analysis based on the methodological description provided in the article and any supplementary materials.

Reproducers were asked to classify computational reproducibility into three categories: *fully reproducible* when the reproduced estimates match the reported results or almost fully reproducible within minor numerical differences attributable to software versions, rounding, or computational precision; *partially reproducible* when some, but not all, of the main claims could be reproduced—whether because some studies within a multi-study article reproduced while others did not, or because, within a given result, some reported quantities reproduced (*e.g.*, the direction and significance of an effect) while others did not (*e.g.*, the associated effect size or confidence interval); *not reproducible* when none of the reported results could be obtained using the available data, code, and documentation. These classifications provide a standardized framework for assessing computational reproducibility across the set of reproduced studies.

Coding Errors and Statistical Inconsistencies: Reproduction teams flagged two families of issues, formally defined in Table 3: *coding errors*, meaning mistakes in the analysis code itself, and *statistical reporting and documentation inconsistencies*, meaning inaccuracies in the reported results or unclear documentation. These issues emerged when teams attempted to execute the original analysis code or when they independently reconstructed parts of the analysis using the original data; however, they exclude very minor issues such as versioning problems, missing packages, and changed file paths. Teams described the coding errors and other issues they encountered in free text; from these descriptions, we constructed the following indicators. We grouped the identified issues into seven categories, formally defined in Table 3: (1) Data construction and preprocessing errors refer to mistakes in transforming, recoding, or combining variables, including incorrect dataset merging or scale construction; (2) coding implementation errors capture general programming mistakes, such as misnamed or outdated variables, incorrect object references, wrong weights, or undefined elements in the code; (3) model specification errors arise when the statistical model implemented differs from the intended analysis, including incorrect model choices or omitted parameters; (4) data filtering and sample construction errors involve mistakes in defining the analytic sample, such as incorrect inclusion or exclusion of observations or missing filtering steps; (5) reproducibility or execution failures occur when code cannot be successfully run due to missing files or other technical issues; (6) documentation and transparency issues capture cases where code is unclear, insufficiently annotated, or ambiguous in its use of data; and (7) statistical reporting errors refer to inaccuracies in reported results, including incorrect summary statistics, p-values, confidence intervals, or inconsistencies between tables and text; number rounding is excluded from this category unless the rounding is incorrectly performed. Categories (1)–(5) constitute coding errors, while categories (6) and (7) represent broader reporting and documentation issues, *i.e.*, statistical inconsistencies/discrepancies. Articles could contain multiple issues across categories.

We also classified the severity of each discrepancy. We coded an issue as *minor* when it is unlikely to affect the interpretation of the findings, for example, a reproducibility obstacle involving secondary parameters, or one that is readily solvable, and as *major*: when it has potential implications for estimates, standard errors, statistical significance, or interpretation, including reproducibility obstacles that involve primary parameters or that cannot be resolved. Table 4 shows examples of the reported issues along with their classifications.

Deviations from Preregistration: We also evaluate whether published analyses deviated from preregistered research plans. Reproduction teams reported whether any of the article's studies was preregistered, whether the original authors deviated from their preregistration, and, in the case of deviations, whether these deviations were disclosed. Teams also described the deviations in detail. Based on these descriptions, we classified each into one of five categories, formally defined in Table 5: Estimation/model; variable construction/measurement; sample size and data collection; data exclusion/filtering; and design and reporting scope. Table 6 provides examples of these deviation issues and their classifications. We also recorded the reason each deviation arose, distinguishing three cases: information specified in the pre-analysis plan (PAP) is omitted from the article; information reported in the article was not prespecified in the PAP; or a procedure specified in both the PAP and the article is implemented differently from what was originally described. We further classified deviations by their implications. We define *major* deviations as departures from the preregistered plan that could plausibly affect the study's results, statistical conclusions, interpretation, or credibility; these include changes to the main analysis, model specification, sample construction, outcome variables, or (non-)reporting of preregistered analyses. In contrast, *minor* deviations refer to changes that are unlikely to meaningfully affect the results or interpretation of the study, such as procedural adjustments, small technical corrections, or analytic changes that leave the substantive conclusions unchanged. Lastly, reproducers reported in the coauthor survey which components of the preregistration were specified: hypotheses; sample size or power calculation; inclusion/exclusion criteria; variable construction; primary outcome variables; primary predictor or treatment variable; other control variables; statistical model or estimation strategy; multiple-hypothesis correction; robustness checks; subgroup or heterogeneity analyses; and data-cleaning procedures.[4]

Perspectives on Preregistration: In addition to the article-level measures above, the coauthor survey elicited each reproducer's attitudes toward preregistration and their personal experience with it. We use four items. First, reproducers rated how much preregistration improves the credibility of research findings on a 0–100 scale, where 0 denotes no improvement and 100 denotes that it strongly improves credibility. Second, they reported how likely they believe a preregistered study

[4]This list is broader than, but encompasses, the six core elements that Hahn et al. (2025) identify as constituting a complete preregistration for confirmatory research (the hypothesized pattern of results, measures, sample size, exclusion criteria, planned analyses, and a time stamp). The only exception being that our survey did not separately record the presence of a time stamp.

is to replicate in new data compared to a non-preregistered study, on a five-point ordinal scale ("unlikely," "somewhat unlikely," "neither likely nor unlikely," "somewhat likely," "likely"). Third, for articles that were preregistered, reproducers indicated whether its presence made reproducing the results easier, on a scale from "much harder" through "no difference" to "much easier," with a "not applicable" option for articles without a plan. We report the distributions of these items in Section 4.5 and relate them to the computational-reproducibility outcomes of the assigned articles to assess whether reproducer attitudes are associated with the outcomes they report. To probe whether reproducer attitudes were associated with the outcomes they reported, we linked each team's mean attitudes to the computational-reproducibility outcome of its assigned article, coded as an indicator equal to 1 for full reproducibility and 0 otherwise.

4 Results

4.1 Data and Code Availability

The availability rates of research materials are summarized in Table 7. This table and our sample excludes unreproduced articles published in 2024—of which about 48.718% lacked publicly available data or code. The findings on computational reproducibility therefore reflect the majority of articles with accessible data and code, but exclude many which lack these resources. See Figure 1.

Availability was highest for downstream materials, that is, the final data and analysis scripts. Final analysis datasets were fully available for 74.6% of articles, partially available for 17.9%, and unavailable for only 7.5%. Likewise, analysis scripts were fully available for 70.1% of articles, partially available for 25.4%, and unavailable for 4.5%. However, upstream materials were shared less often. Raw data were fully available for 46.3% of articles, partially available for 25.4%, and unavailable for 28.4%. Similarly, data-cleaning scripts were fully available for 46.3% of articles, partially available for 23.9%, and unavailable for 29.9%.

4.2 Computational Reproducibility

The results of computational reproducibility from raw or final data are summarized in Table 7. Reproducibility was considerably higher from final data. Beginning from final analysis datasets, 58.2% of articles were fully reproducible, 32.8% partially reproducible, and 9.0% not reproducible. In contrast, beginning from raw data, 28.4% of articles were fully reproducible, 34.3% partially reproducible, and 37.3% not reproducible. While all teams attempted to computationally reproduce the results using the same programming language as the original authors, we note that 25 teams re-coded the analyses using different software.

Table 8 compares articles for which a team re-coded the analysis in different software with those for which teams only re-ran the original code. Reproducibility from final data was broadly similar across the two groups, and slightly higher when the analysis was re-coded in a different software (60.0% fully reproducible, compared with 57.1% without re-coding). However, this difference (2.9 percentage points) was not statistically significant (Fisher's exact test, $p = 1.000$; 95%

CI [-21.490, 27.204]). Our reproducibility rates therefore do not merely reflect re-execution of the original code: results tend to be computationally reproducible when the codes are rewritten in a different language. Reproducibility from raw data was lower when teams re-coded in a different software (20.0% compared with 33.3%), consistent with the greater difficulty of reconstructing the full pipeline in another language. This difference in reproducibility from raw data was likewise not statistically significant (Fisher's exact test, $p = 0.277$; 95% CI [-34.526, 7.859]). Because teams were not randomly assigned to recode the analysis but chose to do so themselves, the recoding and non-recoding groups may differ systematically in ways that also affect reproducibility, for example, in the skill or experience of the team, or in the characteristics of the articles they selected. Any difference in reproducibility between the two groups may therefore reflect these underlying differences rather than the act of recoding itself, so we interpret the comparison as descriptive rather than causal.

Reproducibility also varied with the quality of documentation. As shown in Table 9, packages with a clear README were fully reproducible from final analysis data far more often (87.5%) than those with an unclear one (48.3%), with a difference of 39.224 percentage points (Fisher's exact test, $p = 0.012$; 95% CI [14.865, 63.584]). An unclear README, however, seems to offer little advantage over no README (50.0%) for full reproducibility from analysis data (difference = -1.724 percentage points, Fisher's exact test, $p = 1.000$; 95% CI [-29.425, 25.976]), signaling that it is the clarity of documentation rather than its mere presence that is associated with reproducibility. For computational reproducibility from raw data, the full reproducibility rate was a little higher under a clear README (31.3%) than under an unclear (27.6%) or absent (27.3%) one. None of these differences was statistically significant. These figures are descriptive and based on small subsamples, and clear documentation likely proxies for more careful coding and data-management practices.

4.3 Coding Errors and Statistical Inconsistencies

Reproduction teams identified coding errors or statistical reporting discrepancies in 62.687% of the reproduced articles, a total of 42 articles. The distribution of coding and statistical issues is summarized in Table 10. Among articles with at least one identified issue, statistical reporting errors were the most common (38.1%), closely followed by coding/implementation errors (33.3%). Data filtering/sample construction errors and reproducibility/execution failures were next (23.8% each), and documentation and transparency issues affected 19.0% of these articles. Data construction and preprocessing errors were the least common (2.4%). Across all reproduced articles, coding errors were present in 43.284% and statistical reporting and documentation inconsistencies in 32.836%.

The identified coding and statistical issues varied in their implications for the validity of original claims. More than half of the articles flagged with discrepancies (59.524%) contained only minor issues that unlikely affect the interpretation of the findings. In contrast, 40.476% of these articles contained issues with potentially major implications for estimates, standard errors, statistical significance, or interpretation.

4.4 Deviations from Preregistration

The results are summarized in Table 11. Preregistration was common among the articles in our sample. Reproducers reported that 74.6% of articles (50 of 67) included at least one preregistered study, while 25.4% (17 articles) contained no preregistered study. Among preregistered studies, core design elements were almost universally specified. Figure 3 reports the prevalence of different components included in pre-analysis plans, as documented by the reproducers. More than 90% of preregistrations include information on statistical models or estimation strategies, sample size or power calculations, primary outcome variables, hypotheses, primary predictor or treatment variables, and inclusion or exclusion criteria. Variable construction is also commonly documented, appearing in approximately 80% of preregistrations. However, other components were less frequently specified: control variables appeared in slightly more than half of plans, data-cleaning procedures in fewer than half, robustness checks and subgroup or heterogeneity analyses in a minority, and multiple-hypothesis correction in fewer than 10%. More than 80% of the preregistrations were hosted on the Open Science Framework, with the remainder on AsPredicted.

Among the 50 preregistered reported articles, 42 (84%) deviated from the pre-analysis plan in at least one respect, and 8 (16%) did not. Of the 42 articles with deviations, 25 (59.5%) disclosed some or all of them, while 17 (40.5%) disclosed none. The most common form of deviation was implementing a different procedure in the paper from the preregistration, occurring in 57.143% of preregistered articles with deviations. The prevalence of these categories is summarized in Table 12. Deviations involving design or reporting scope were most frequent, occurring in 25 articles, followed by estimation or modeling choices, occurring in 23 articles. Deviations related to sample size/data collection and data exclusions/filtering were each identified in 12 article, while deviations concerning variable construction were prevalent in 10.

Turning to the implications of these deviations, we find that both major and minor deviations were prevalent among the reproduced articles; each of these deviations equally appeared in 28 articles. Since articles can contain multiple deviations spanning different categories, these counts are not mutually exclusive and therefore exceed the 42 articles with identified deviations.

4.5 Perspectives on Preregistration

Most reproducers report that preregistration made the reproduction of their assigned article easier (Figure 4). As reported in Table 1, reproducers express generally favorable views toward preregistration, assigning an average score of 69.8 out of 100 to the statement that preregistration improves research credibility. The majority of respondents also believe that findings from preregistered studies are more likely to replicate in independent samples, as shown in Figure 5. These attitudes are positively correlated: reproducers who believe preregistration improves credibility also tend to believe a preregistered study is more likely to replicate (Figure 6). Nonetheless, a reproducer's experience with preregistration appears to be unrelated to the belief that preregistration improves the credibility of research findings. As illustrated in Figure 7, the correlation between the two measures is small and statistically insignificant ($r = 0.074$; df = 223; 95% CI $[-0.057, 0.203]$; $p = 0.268$).

Beyond how these attitudes relate to one another, we also examined whether they relate to the reproduction outcomes themselves. Full reproducibility from raw data was unrelated to both the team's rating of whether preregistration improves credibility ($r = 0.112, p = 0.369$) and its level of personal preregistration experience ($r = -0.030, p = 0.807$). For full reproducibility from analysis data, the associations were stronger but did not reach significance ($r = 0.227, p = 0.065$, and ($r = -0.215, p = 0.081$). Given the small number of reports and the limited range of attitudes in our pool, we treat these patterns as exploratory.

5 Discussion

This project provides a comprehensive assessment of computational reproducibility in *Psychological Science*, drawing on 67 independent reproduction reports that cover 64 (44%) of research articles published in the journal between 2024 and 2025. Three broad patterns emerge within our sample. First, transparency practices are now widespread, but they thin out as one moves upstream from final analyses toward raw data. Second, computational reproducibility is high when teams begin from final analysis datasets and substantially lower when they attempt to rebuild the analytical workflow from raw data. Third, coding errors, statistical reporting inconsistencies, and undisclosed deviations from preregistered plans remain common, although most of the issues we identified are unlikely to overturn the original conclusions. We discuss each of these patterns next, and then consider the implications, limitations, and future directions of the initiative.

The availability rates we observe are consistent with the adoption of open-data and open-code practices in the journal (Hardwicke and Vazire, 2024). However, availability is markedly uneven across the workflow: final datasets and analysis scripts were fully available for the large majority of articles, whereas raw data and cleaning scripts were fully available for fewer than half. This gradient matters because the upstream materials are precisely those needed to reconstruct how an analysis sample was built from the original observations. The state of documentation reinforces the concern: Less than a quarter of replication packages included a README that reproducers judged clear, while roughly a third included none at all. This matters because documentation quality was closely tied to reproduction success: packages with a clear README were fully reproducible from final analysis data far more often than those with an unclear README or none at all. Notably, an unclear README offered little advantage over no README, indicating that it is the clarity of documentation, rather than its mere presence, that facilitates reproducibility. Inadequate documentation therefore remains a common obstacle to efficient computational reproduction, even when data and code are technically available.

The starting point of a reproduction attempt strongly conditions whether it succeeds. A majority of articles were fully reproducible from final analysis datasets, but only about three in ten were fully reproducible from raw data, with the remainder split between partial and failed reproductions. The size of this gap suggests that reconstructing the data-cleaning and sample-construction pipeline, not re-running the final models is the binding constraint on reproducibility. Sharing a clean final dataset and an analysis script does most of the work required to verify the headline numbers, but it leaves

the steps that transform raw observations into the analytic sample opaque. This limits the ability of independent researchers to conduct sensitivity analyses or to detect errors introduced before the final dataset is built, and it helps explain why the availability of raw materials and reproducibility from raw data move together.

Reproduction teams identified at least one coding error or statistical reporting discrepancy in a majority of articles, with statistical reporting errors and coding implementation errors the most frequent types. Nonetheless, severity was generally low: more than half of flagged articles contained only minor discrepancies, while the rest contained issues with potentially major implications for estimates, statistical significance, or interpretation. Most coding and statistical inconsistencies are therefore unlikely to alter the substantive conclusions of the original studies, although a non-trivial minority could. These rates are difficult to benchmark precisely, but they are broadly in line with prior work. We identified coding errors and statistical reporting/documentation inconsistencies in 43.3% and 32.8% of our articles, respectively. By comparison, Brodeur et al. (2026b) documented coding errors in about 25% of articles in leading economics and political science journals, while Nuijten et al. (2016) found that roughly half of published articles in psychology contained at least one inconsistency.

Preregistration has become close to standard practice, and preregistrations almost universally specified the core design. Preregistrations thus tend to provide detailed information on the primary design and analysis of a study but leave secondary and procedural choices unspecified. The majority of the preregistered articles contain some deviation, and these deviations were frequently undisclosed. Our results are consistent with prior work on preregistration adherence in psychology. Claesen et al. (2021), examining an earlier cohort of preregistered studies in *Psychological Science*, found deviations from the prespecified statistical analysis in 67% of cases, and van den Akker et al. (2024) found that the choice of statistical model was a source of plan–study discrepancy in about 75% of cases. Because inclusion in our sample required a replication package, our estimates should be compared with this literature only with that selection in mind. Our analyses thus contribute to a growing literature documenting the limited clarity of pre-analysis plans in the social sciences and the prevalence of undocumented deviations from preregistration (Brodeur et al., 2024; Franco et al., 2016; Ofosu and Posner, 2023). The prevalence of deviations also raises the question of how best to characterize them. Our classification captures which aspect of the analysis changed rather than why the deviation occurred. A complementary approach, such as the taxonomy proposed by Lakens (2024) instead classifies deviations by their underlying reason—unforeseen events, mistakes in the preregistration, missing information, violations of untested assumptions, or the falsification of auxiliary hypotheses—which in turn bears on whether a deviation reduces or preserves the severity of a test. Applying such a reason-based classification reliably requires the authors' own stated justifications, however, which were frequently unavailable in our setting given that a substantial share of deviations were undisclosed. We therefore retain a structural categorization and leave reason-based coding to future work with access to author-reported rationales.

The reproducer survey adds a complementary perspective. Reproducers held generally favorable views of preregistration, rating its contribution to credibility highly and generally expecting

preregistered findings to replicate. Notably, reproducers' own preregistration experience was unrelated to the belief that preregistration improves credibility. This contrasts with Spitzer and Mueller (2023), who reported that researchers with more preregistration experience held more positive views. A plausible explanation is a ceiling effect: volunteers at a reproducibility initiative may already hold favorable views of open science practices, irrespective of their personal experience. This is consistent with Field et al. (2020) who found little evidence linking trust in research to preregistration in a broader sample.

5.1 Limitations

None of our analyses were preregistered, and they should be viewed as descriptive and exploratory. The patterns we document—rates of computational reproducibility, coding errors, and deviations from preregistration—should be understood as characterizations of current practice in a specific journal over a specific time window, and do not constitute claims about discipline-wide reproducibility.

Two features of our design further suggest that the estimates reported here are best read as optimistic bounds. First, reproducers self-selected into the initiative and, as documented in Section 4.5, tended to hold favorable views of open science practices, so their attitudes should not be taken as representative of the broader research community. Second, and more consequentially for our reproducibility and error estimates, teams chose which articles to reproduce and explicitly avoided studies with very large datasets, long runtimes, unfamiliar methods, or incomplete data. Roughly, 20% of otherwise eligible articles published in 2024 shared neither data or code. If more complex studies are both more error-prone and harder to reproduce, our sample is drawn disproportionately from the simpler, more tractable tail of the literature. Under that assumption, the coding errors and statistical inconsistencies we document represent a lower bound on their true prevalence, while our computational reproducibility rates represent an upper bound on what independent verification would recover across all published articles.

Lastly, around 84.4% of the reproduced articles were submitted before the arrival of the current editorial board. Therefore, most of these articles were not subject to the journal's strengthened transparency policies and do not constitute an assessment of the journal's current standards. We view this study as a baseline analysis: it sets a starting point that future reproductions can be compared to. Our next target is precisely that question: whether the journal's strengthened expectations regarding data, code, and materials sharing translate into measurable improvements in transparency and reproducibility. Future waves of reproductions, covering articles submitted under the new policy, will allow us to move toward a thorough assessment of the policies' effects.

5.2 Recommendations

Our reproduction teams encountered cases where packages appeared to have been altered post-publication, making it impossible to determine whether discrepancies arose from coding errors in the original submission or from subsequent undocumented edits. Currently, the vast major-

ity of replication materials in our sample were deposited on the Open Science Framework (OSF). While OSF does support a registration-based archival mechanism, which prevents subsequent edits and has been used successfully by other journals (*e.g.*, Meta-Psychology, 2026), this functionality is not currently required by *Psychological Science*, and OSF's standard project storage remains editable indefinitely absent such a registration. Moreover, in August 2026 the Center for Open Science announced that it will discontinue OSF's general-purpose project storage and collaboration functionality (Center for Open Science, 2026). This transition reinforces the need for a dedicated, purpose-built archival solution. We recommend that *Psychological Science* require replication packages to be deposited in repositories with stronger archival guarantees and immutable versioning, such as the Harvard Dataverse, Zenodo, or ICPSR (Inter-university Consortium for Political and Social Research). Note that Zenodo deposits remain editable for 30 days after upload before becoming permanently immutable, so authors should be advised to finalize materials before this window closes. These platforms provide persistent, citable, and timestamped records of deposited materials, ensuring that the package a reproduction team accesses years after publication is identical to what was available at the time of publication. Another option is ResearchBox, a platform tightly integrated with AsPredicted, allowing preregistrations to be imported seamlessly alongside deposited data and code. More broadly, ensuring archival standards such as these are consistently adopted and enforced would benefit from dedicated editorial oversight; we therefore also recommend that the American Psychological Association (APA) introduce a dedicated data editor role for all its journals, as has already been done by scientific associations in economics and political science, for instance.

A second recommendation concerns the timing of data and code sharing in the review process. Our results show that coding errors and statistical reporting inconsistencies reached publication without detection, suggesting that peer review as currently practiced offers limited protection against such problems. Requiring authors to submit their replication package at the revision stage (R&R) would create a natural checkpoint at which reviewers could verify that reported results are computationally reproducible before a manuscript is accepted. This need not impose a heavy burden on authors as this could be required only for articles that receive a R&R decision. The marginal cost of earlier sharing is likely low, while the potential gains in accuracy and transparency are substantial. The extra time devoted to checking the replication package for reviewers at the R&R stage would not be too onerous as even a cursory check that code runs and produces the reported estimates would catch the most consequential errors documented in our sample. Recent experimental evidence suggests AI tools can support checking computational reproducibility specifically, without sacrificing reliability, as long as they complement rather than replace human involvement (Brodeur et al., 2026a).

A third recommendation concerns researcher training in data management and scientific computing. The high proportion of coding errors found in our sample points to potential gaps in computational training that psychologists receive. At the graduate level, training programs should incorporate dedicated instruction in reproducible research practices, including version control, code documentation, and data management, as core methodological competencies. For established re-

searchers, professional societies are well positioned to reach this group through workshops and mentorship programs pairing computationally experienced collaborators with those seeking to strengthen their practices. Another potential approach is the adoption of a red team logic that integrates criticism directly into the research process: one team conducts the research while an independent team actively attempts to identify errors, challenge assumptions, and probe the robustness of the findings at each stage. Embedding collaboration of this kind into standard research workflows could catch the kinds of coding and analytical errors documented in our sample before they reach publication.

A fourth recommendation concerns the quality and availability of README documentation accompanying replication packages. Our findings reveal that a minority of replication packages in our sample included a clear README file. A well-written README is arguably the single most important component of a replication package: it orients an independent reproducer to the structure of the package, maps data files to analysis scripts, documents software dependencies and version requirements, and specifies the order in which code should be executed to reproduce each table and figure in the article. It should also specify computational requirements in full: the software and package versions used, the hardware specification, and the wall-clock runtime required to reproduce the results. Our own data illustrate why this matters— *e.g.,* reported runtimes in our sample ranged from under five minutes to 40 hours, variation that a reproducer cannot anticipate without this information being documented upfront. Without it, even a complete and error-free replication package can become nearly impenetrable to an outside team, as our reproducers frequently discovered. Improving README quality across the discipline does not require reinventing the wheel. For example, the social science data editors community has developed a widely adopted template README for replication packages that provides clear guidance on the information that should be included and how it should be organized (Vilhuber et al., 2022). Psychological journals would do well to adopt such a template as a requirement, ensuring that replication packages are navigable and that the investment authors make in sharing their materials translates into genuine reproducibility.

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Figures and Tables

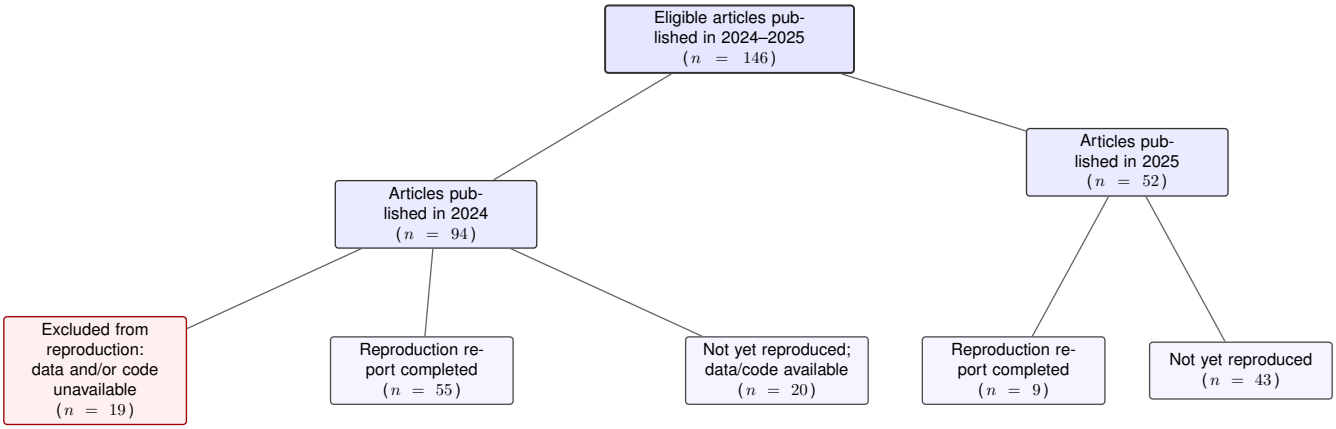


Figure 1: Flow of Eligible Articles through the Reproduction Process

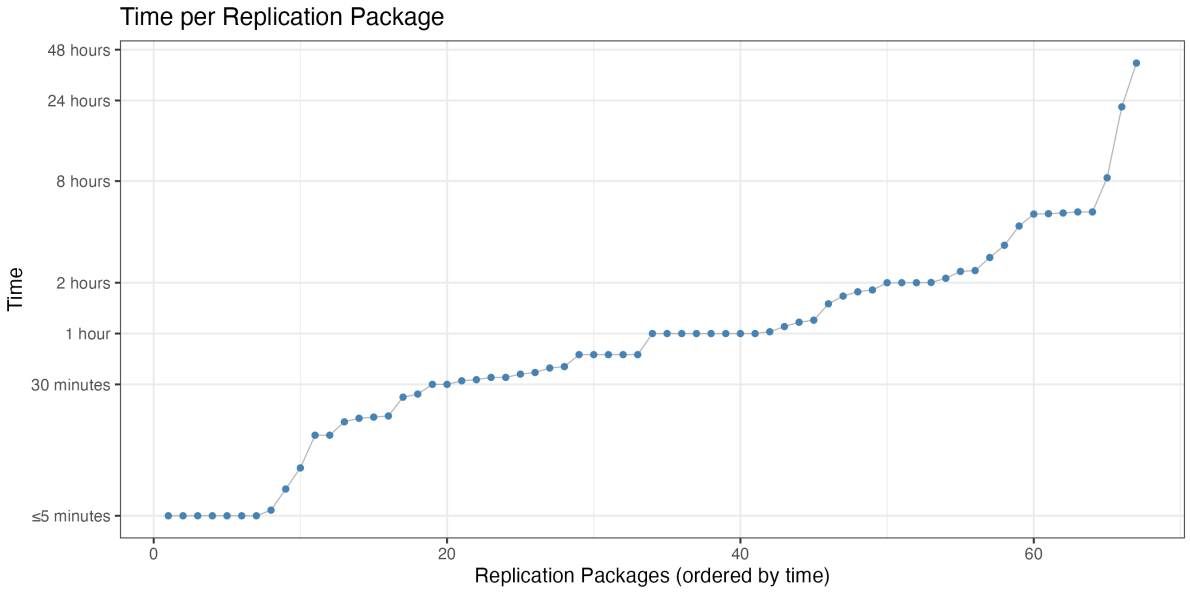


Figure 2: Runtime of the Original Authors' Code

Note: Each point represents one replication package. Packages are ordered from the shortest to the longest reported runtime. By runtime we mean how many hours it takes for all the codes to run, from initial cleaning until all results are completed, excluding time spent fixing the code to make it run. A nonlinear transformation was applied to the y-axis to improve readability across the range of times.

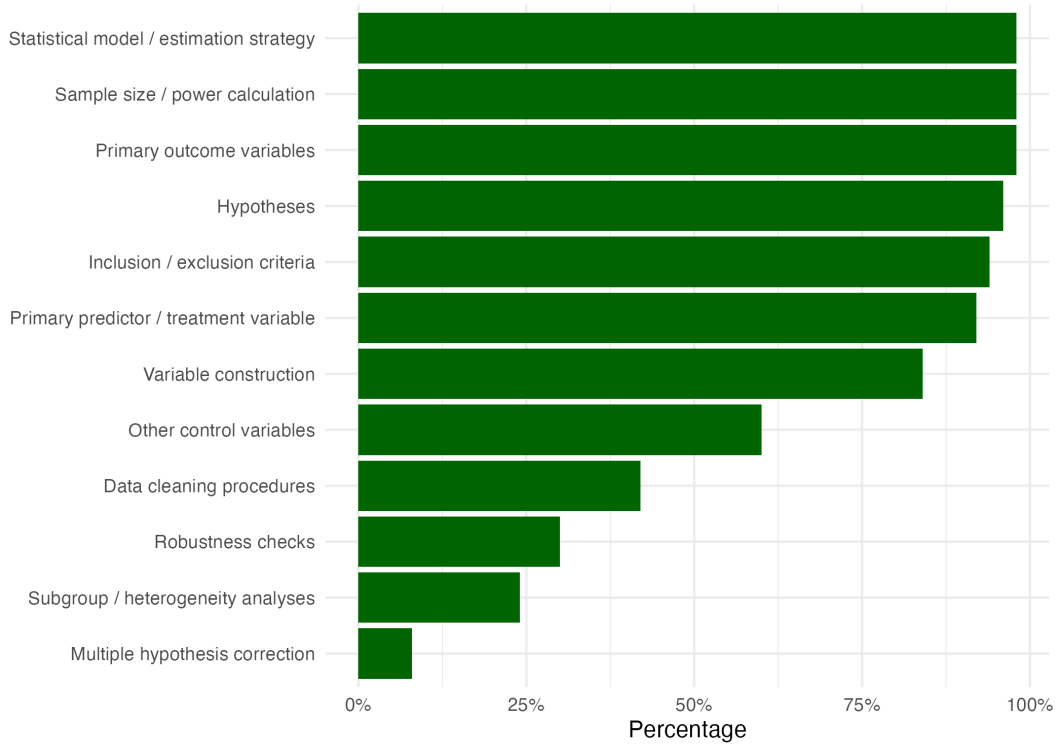


Figure 3: Specified Elements of the Pre-analysis Plan

Note: This figure shows the percentage of reproducers reporting that each component was included in their pre-analysis plan (PAP). Reproducers could select multiple components.

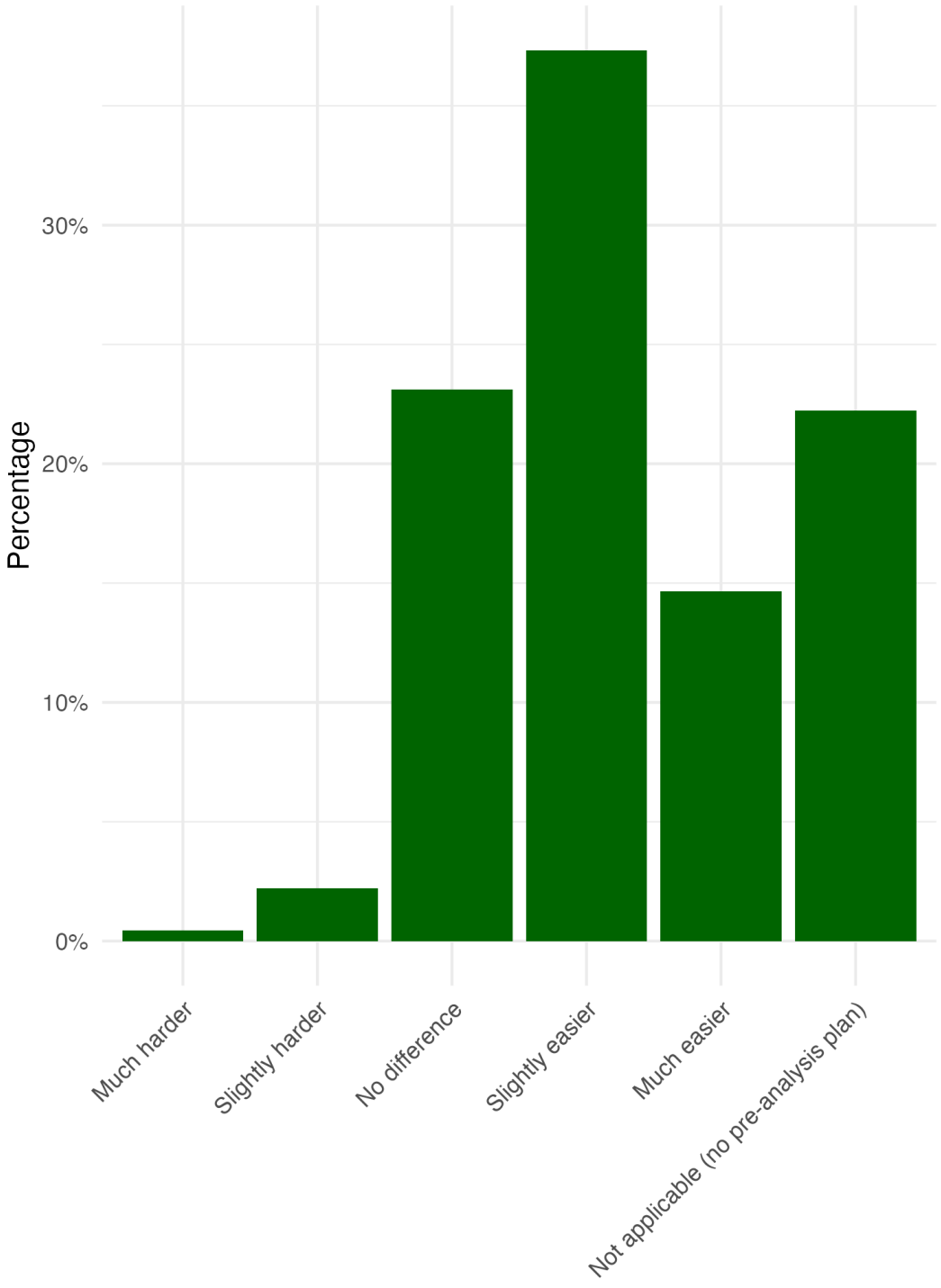


Figure 4: Perceived Effect of Pre-Analysis Plans on Reproduction Ease

Note: This figure displays the distribution of responses for the following question: Did the presence of a pre-analysis plan make reproducing the results of this paper easier?

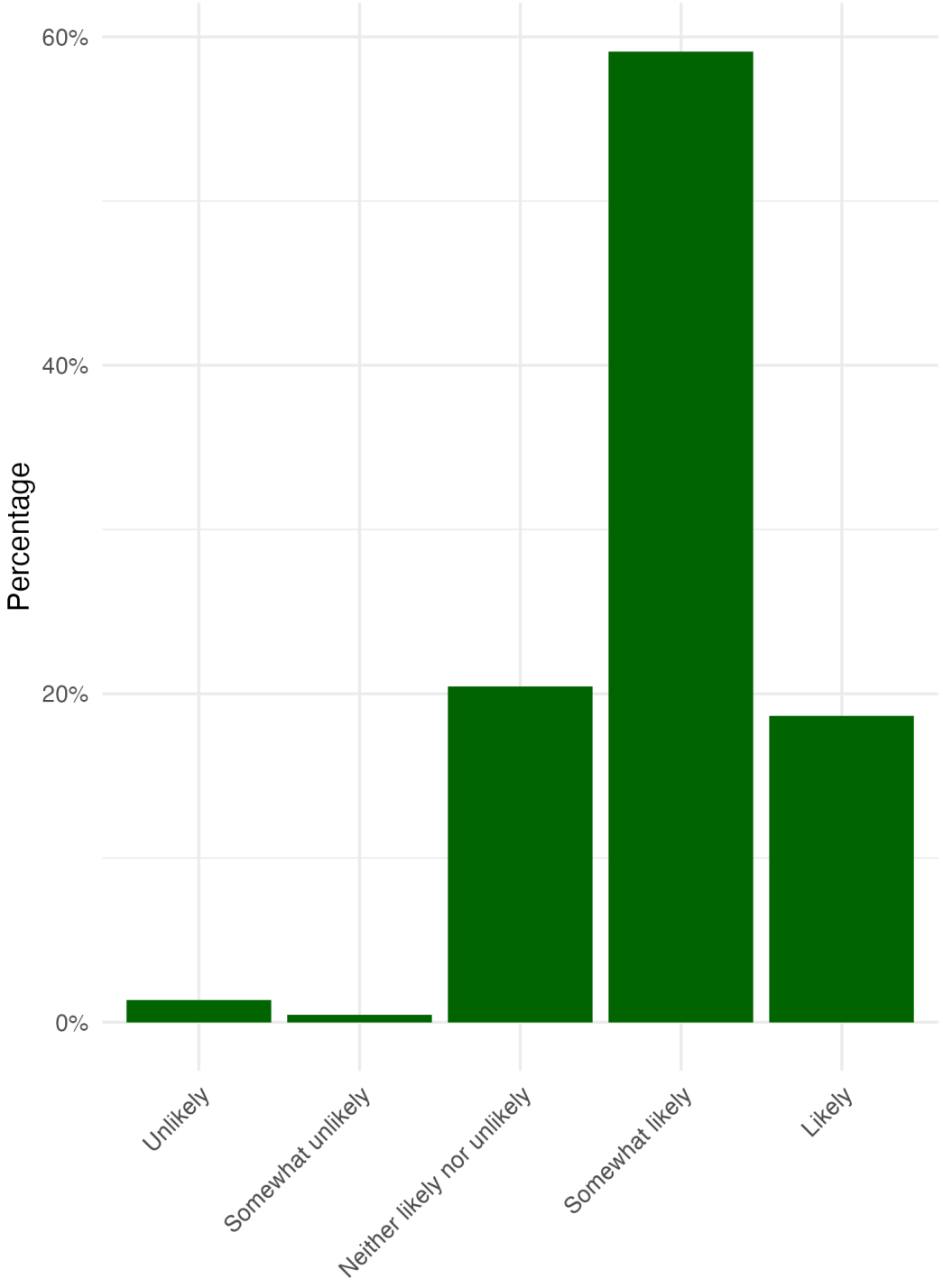


Figure 5: Perceived Replication Likelihood of Preregistered Studies

Note: This figure displays the distribution of responses for the following question: How likely do you believe a preregistered study is to replicate (i.e., using new data) compared to a non-preregistered study?

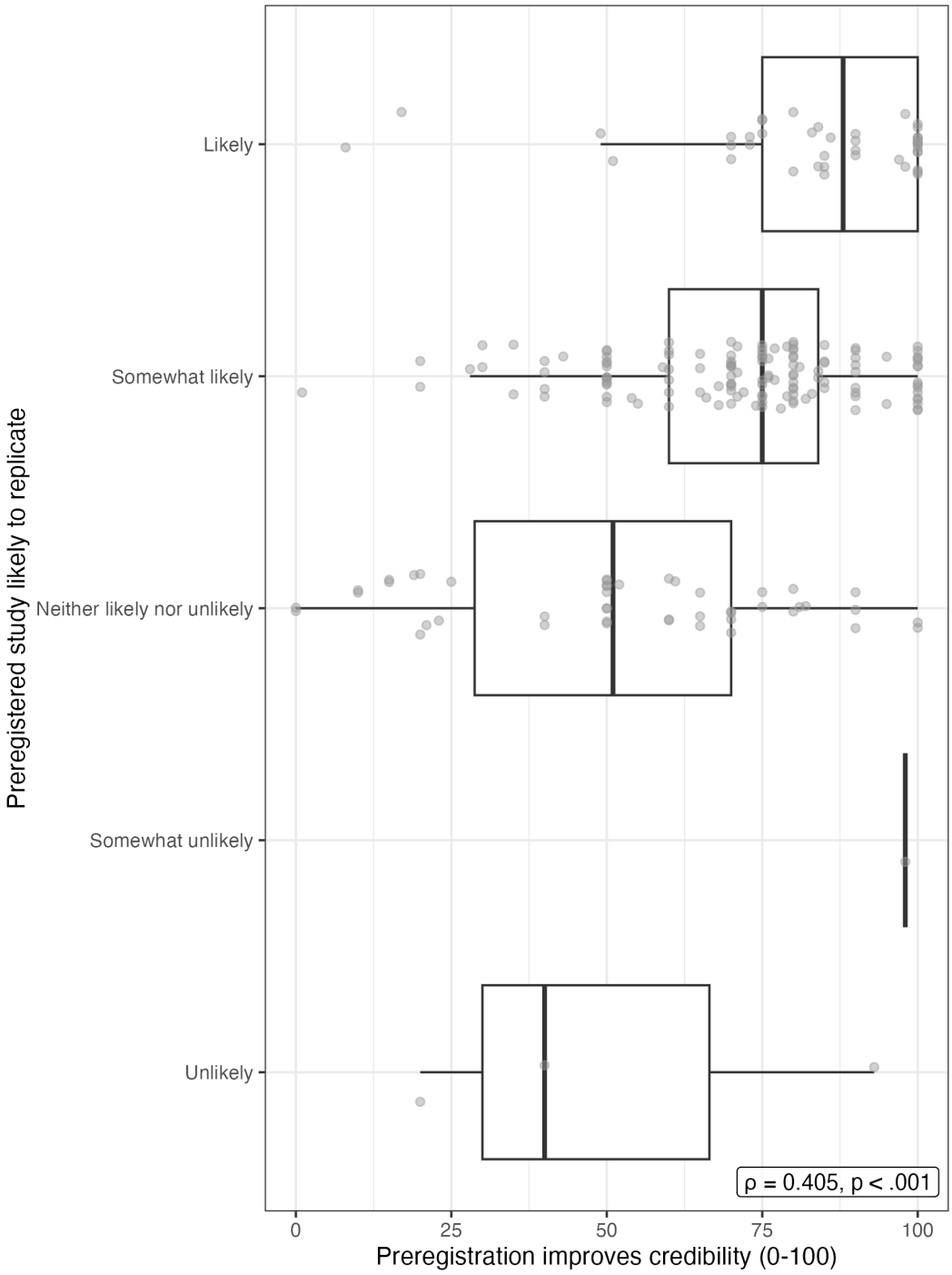


Figure 6: Relation between beliefs on the impact of preregistration for credibility and the likelihood of replication.

Note: Points represent individual responses. The Spearman rank correlation and its significance is displayed in the lower right.

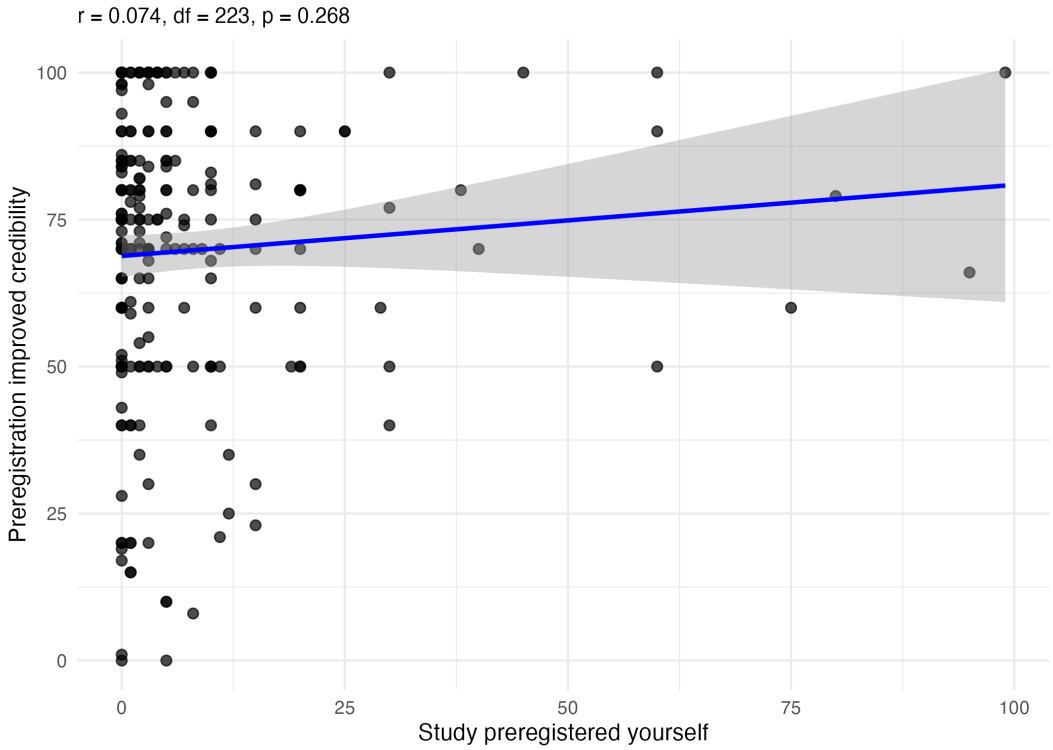


Figure 7: Correlation between preregistration experience and beliefs on the impact of preregistration for credibility.

Note: This figure displays the Pearson correlation between the responses to the following answers. x-axis: Approximately how many studies have you personally preregistered? y-axis: In your opinion, how much does preregistration improve the credibility of research findings? 0 = No improvement; 100 = Strongly improves credibility. The shaded band around the fitted line represents the 95% confidence interval of the linear fit.

Table 1: Summary of Original Author and Reproducer Characteristics

Variable	Mean	SD	Min	Max
Panel A: Original Author Characteristics				
Graduate student	0.139	0.347	0.000	1.000
Postdoctoral fellow	0.124	0.330	0.000	1.000
Assistant professor	0.180	0.385	0.000	1.000
Associate professor	0.105	0.307	0.000	1.000
Professor	0.282	0.451	0.000	1.000
Researcher	0.120	0.326	0.000	1.000
Other	0.049	0.216	0.000	1.000
Google Scholar citations	6618.752	18174.553	0.000	163422.000
Panel B: Reproducer Characteristics				
Graduate student	0.507	0.501	0.000	1.000
Postdoctoral fellow	0.182	0.387	0.000	1.000
Assistant professor	0.102	0.304	0.000	1.000
Associate professor	0.044	0.207	0.000	1.000
Professor	0.053	0.225	0.000	1.000
Researcher	0.076	0.265	0.000	1.000
Other	0.036	0.186	0.000	1.000
Google Scholar citations	1363.361	4414.611	0.000	33771.000
Number of years programming	8.058	7.302	0.000	45.000
Number of days working on reproduction	5.773	4.652	1.000	30.000
Psychology field	0.711	0.454	0.000	1.000
Produced replication package	0.387	0.488	0.000	1.000
Number of preregistered studies	7.764	14.746	0.000	99.000
Preregistration improves credibility	69.751	24.033	0.000	100.000

Note: Column 1-4 report the mean, standard deviation, minimum and maximum, respectively. Position rows treat each position category as a 0/1 dummy variable. Graduate students include masters and Ph.D. students. Other category includes undergraduate students or those whose current position is unknown. Psychology field equals 1 if the reported field is psychology-related and 0 otherwise. Number of days working on reproduction reflects reproducers' self-reported count of active days spent on their assigned report, from assignment until publication as a discussion paper. Produced replication package equals 1 for yes and 0 for no. Preregistration improves credibility is measured on a 0–100 scale. Total number of authors is 266. Total number of reproducers is 225.

Table 2: Summary of Article and Replication Package Characteristics

Variable	Mean	SD	Min	Max
Number of authors per article	4.156	5.056	1.000	41.000
Number of reproducers per article	3.358	1.276	1.000	6.000
Number of studies per article	3.433	2.457	1.000	11.000
R coding language (authors)	0.836	0.373	0.000	1.000
R coding language (replicators)	0.896	0.308	0.000	1.000
Computational time	2.287	5.558	0.000	40.000
Clear README	0.239	0.430	0.000	1.000
Unclear README	0.433	0.499	0.000	1.000
Unprovided README	0.328	0.473	0.000	1.000

Note: R coding language rows are coded as 1 if R was used and 0 otherwise. Computational time corresponds to the number of hours it takes to run the codes provided by the original authors from initial cleaning until all results are completed. README rows are dummy-variable summaries for each README file category. Unprovided README combines Not Provided and Not Applicable.

Table 3: Categories of Coding Errors and Statistical Reporting Inconsistencies

Category	Definition
Panel A: Coding Errors	
Data construction / preprocessing	Mistakes in transforming, recoding, or combining variables, including incorrect dataset merging or scale construction, and preprocessing pipeline mistakes.
Coding / implementation	General programming mistakes, such as misnamed or outdated variables, incorrect object references, or undefined elements in the code.
Model specification	Cases where the implemented statistical model differs from the intended analysis, including incorrect model choices, omitted parameters, or wrong weights.
Data filtering / sample construction	Mistakes in defining the analytic sample, such as incorrect inclusion or exclusion of observations or missing filtering steps.
Reproducibility / execution failure	Code that cannot be run successfully due to missing files or other technical issues preventing the code from running.
Panel B: Other Issues	
Documentation / transparency	Code that is unclear, insufficiently annotated, or ambiguous in its use of data, or poor variable naming that hinders interpretation.
Statistical reporting	Inaccuracies in reported results, including incorrect summary statistics, p-values, or confidence intervals, or inconsistencies between tables and text; number rounding is excluded unless the rounding is performed incorrectly, along with truncation and sign errors.

Note: Categories in Panel A constitute **coding errors**; categories in Panel B constitute **statistical reporting and documentation inconsistencies**. An article may contain multiple issues spanning several categories, so the categories are not mutually exclusive.

Table 4: Examples of Error Classifications

Issue	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
incorrect or outdated variable names. We found errors in pre-processing of the stigma consciousness scale (Pinel, 1999). Questions were asked using different scale numbers and errors were found in which items were reverse coded. Nevertheless, this variable was not used in any of the analyses, making these errors inconsequential for the results.	1	1	0	0	0	0	0	1	0
when looking at the age variable, we find a wide range of variation from 3 years old to 80 years old, which is beyond that reported in the paper ("ranging in age from 18 to 76 years"). We do not think this would meaningfully affect the results	0	0	0	1	0	0	0	1	0
When starting the analyses, one of the issues we observed was that 85 participants, who were not part of the sample criteria (i.e., students or healthcare professionals), were not excluded in the beginning of the study. Indeed, when analyzing the data these individuals were categorized as N/A (neither student nor health-care professional). While these individuals were described in Study 1 (descriptives), they were not within the reported results of the remaining analyses. Therefore, the reported results may not have been representative of the sample described and this could be misleading in terms of effect size.	0	0	0	1	0	0	0	0	1
some values had to be changed between Studies 1 and 3, such as the degrees of freedom for the cluster-based permutation t-tests (e.g., 23 for Study 1a, 39 for Study 3a, 19 for Study 3b), but this was not included in the code or specified in the comments. the threshold determination tests on line 1418 the alpha value seemed to have a typo (alpha = .10, rather than potentially .01). data frame names were not consistently updated to reflect removals, edits, and formatting. It was also not obvious which of the datasets was used to calculate inferential statistics such as the t-test and binomial tests	0	1	1	0	0	0	1	1	0
The authors mistakenly reported the lower bound of the 95% confidence interval as the mean difference. At least two of the experiments (Exp.1a and 3a) were analyzed without exclusion of trials based on reaction times, in contrast to what they report in the manuscript	0	0	0	1	0	0	1	0	1
The second model, which included a random slope, produced an error. We were able to troubleshoot convergence-related issues and the model converged. However, the subsequent calculations intended to generate confidence intervals produced an error which we were unable to resolve.	0	0	0	0	0	1	0	0	1
Some minor mistakes could have been found that didn't affect results (e.g., in study 4, the main effect of named voices F statistic, generated with SPSS script, equaled 32.494, and in the manuscript, the reported value is rounded to 32.45, while it should be 32.50).	0	0	0	0	1	0	0	1	0

Note: Columns (2)–(8) present the following categories: Data construction / preprocessing, Coding / implementation, Model specification, Data filtering / sample construction, Reproducibility / execution failure, Documentation / transparency, and Statistical reporting, respectively. Columns (9) and (10) present minor and major severity categories, respectively.

Table 5: Categories of Deviations from Preregistration

Category	Definition
Estimation / model	Changes to the estimation strategy, statistical model, or inferential procedure used to test hypotheses.
Variable construction / measurement	Changes to how variables are defined or built, using different outcome or predictor variables than planned, and recoding variables differently from the preregistration.
Sample size / data collection	Changes to the planned sample size, stopping rule, or data-collection procedure.
Data exclusion / filtering	Changes to the analytic sample through inclusion or exclusion rules, cleaning observations or participants after data collection, or modifying outlier handling.
Design / reporting scope	Changes to the experimental design or to the scope of what is reported, including selective reporting.

Note: An article may contain multiple deviations spanning several categories, so the categories are not mutually exclusive.

Table 6: Examples of Pre-analysis Plan Deviation Categories

Deviation	(2)	(3)	(4)	(5)	(6)	(7)	(8)
there is some theoretical vagueness regarding how their variables were constructed, and they did not provide theoretical justification for the creation of the composite predictor variable scores or why they decided to use difference scores for the prejudice measures. Further, the individual scales that make up the two main predictor composites (anti-Black prejudice and intergroup contact) are not consistent from Study 1 to Study 3. There is no justification provided by the authors in either the main text, the supplemental text, or the pre-registrations for the same. Finally, from the provided code, we see that the main analyses and results reported in the main text were run on a dataset where more outliers were excluded than stated in either the pre-registration or the main text.	0	1	0	1	0	1	0
there were two minor discrepancies, wherein analyses that were not described in the PAP were reported in the published article. First, in the procedure section, YW described a "posttest" to verify that participants viewed the incentives as believable; they provided full details in the supplemental material. The term posttest implies that the analysis was exploratory, though they did not specifically use the term "exploratory". Second, the authors described testing the effect of incentive salience on prosocial motivation. Although the authors referred to this test as exploratory in the supplemental materials, they did not identify it as such in the manuscript.	0	0	0	0	1	1	0
There are also some deviations from the preregistrations. Specifically, the preregistrations for Studies 1b, 1c, and 1d further mention additional hypotheses that have regret as a dependent variable and that are not reported in the main text but are tested in the code or described in the supplementary material. The preregistrations also mention robustness checks without data exclusions, exploratory analyses that are included in the code, and exploratory analyses that have not been conducted—please see Table 5 for details. The pre-registrations of studies 2a and 2b mention English check exploratory analyses which were not included in the code.	0	0	0	0	1	1	0

Note: Columns (2)–(5) present the following categories: Estimation / model, Variable construction / measurement, Sample size / data collection, Data exclusion / filtering, and Design / reporting scope, respectively. Columns (6) and (7) present minor and major severity categories, respectively.

Table 7: Summary of Data, Code Availability, and Reproducibility per Article

	Fully	Partially	No
Raw data	46.269	25.373	28.358
Analysis data	74.627	17.910	7.463
Cleaning code	46.269	23.881	29.851
Analysis code	70.149	25.373	4.478
Reproducible from raw data	28.358	34.328	37.313
Reproducible from analysis data	58.209	32.836	8.955

Note: Entries report percentages of articles "Fully" indicates complete availability/reproducibility, "Partially" indicates incomplete availability/reproducibility, and "No" indicates absence.

Table 8: Summary of Data, Code Availability, and Reproducibility by Re-coding in Different Software Status

	Fully	Partially	No
Panel A: No re-coding (n = 42)			
Raw data	47.619	26.190	26.190
Analysis data	73.810	19.048	7.143
Cleaning code	50.000	21.429	28.571
Analysis code	73.810	23.810	2.381
Reproducible from raw data	33.333	33.333	33.333
Reproducible from analysis data	57.143	33.333	9.524
Panel B: Re-coding (n = 25)			
Raw data	44.000	24.000	32.000
Analysis data	76.000	16.000	8.000
Cleaning code	40.000	28.000	32.000
Analysis code	64.000	28.000	8.000
Reproducible from raw data	20.000	36.000	44.000
Reproducible from analysis data	60.000	32.000	8.000

Note: Entries report conditional percentages within each panel. Panel A includes articles with no re-coding. Panel B includes articles with recoding using different software. "Fully" indicates complete availability/reproducibility, "Partially" indicates incomplete availability/reproducibility, and "No" indicates absence.

Table 9: Summary of Data, Code Availability, and Reproducibility by README Availability

	Fully	Partially	No
Panel A: README not provided (n = 22)			
Raw data	36.364	22.727	40.909
Analysis data	72.727	22.727	4.545
Cleaning code	36.364	22.727	40.909
Analysis code	59.091	31.818	9.091
Reproducible from raw data	27.273	27.273	45.455
Reproducible from analysis data	50.000	45.455	4.545
Panel B: README provided and clear (n = 16)			
Raw data	56.250	18.750	25.000
Analysis data	81.250	12.500	6.250
Cleaning code	50.000	18.750	31.250
Analysis code	81.250	12.500	6.250
Reproducible from raw data	31.250	37.500	31.250
Reproducible from analysis data	87.500	6.250	6.250
Panel C: README provided but unclear (n = 29)			
Raw data	48.276	31.034	20.690
Analysis data	72.414	17.241	10.345
Cleaning code	51.724	27.586	20.690
Analysis code	72.414	27.586	0.000
Reproducible from raw data	27.586	37.931	34.483
Reproducible from analysis data	48.276	37.931	13.793

Note: Entries report conditional percentages within each README availability group. Panel A includes articles where the README was not provided. Panel B includes articles where the README was provided and clear. Panel C includes articles where the README was provided but unclear. "Fully" indicates complete availability/reproducibility, "Partially" indicates incomplete availability/reproducibility, and "No" indicates absence.

Table 10: Summary of Error Categories and Reporting Inconsistencies

Error category	count	percent
Panel A: Coding Errors		
Data construction / preprocessing	1.000	2.381
Coding / implementation	14.000	33.333
Model specification	8.000	19.048
Data filtering / sample construction	10.000	23.810
Reproducibility / execution failure	10.000	23.810
Panel B: Other Issues		
Statistical reporting	16.000	38.095
Documentation / transparency	8.000	19.048

Note: Counts indicate the number of articles with at least one issue in each category. Percentages are calculated relative to the total number of articles with coding errors or reporting inconsistencies (total of 42). Because an article may contain more than one issue that could fall under multiple categories, counts and percentages can sum to more than the total number of articles.

Table 11: Summary of Preregistration and PAP Deviations

	count	percent
Panel A: Preregistration Status		
Preregistered articles	50.000	74.627
Not preregistered articles	17.000	25.373
Panel B: Deviations from Pre-Analysis Plans		
Articles with PAP deviation	42.000	84.000
Articles without PAP deviation	8.000	16.000
Panel C: Deviation Acknowledgment		
Articles with deviation acknowledged	25.000	59.524
Articles with deviation not acknowledged	17.000	40.476

Note: PAP denotes Pre-analysis plan. Percentages for preregistration status are calculated relative to the total number of reproduced articles ($n = 67$). Percentages for PAP deviation status are calculated relative to the number of preregistered articles ($n = 50$). Percentages for deviation acknowledgment are calculated relative to the number of articles with PAP deviations ($n = 42$).

Table 12: Summary of Pre-Analysis Plan Deviation Categories

Deviation category	count	percent
Estimation / model	23.000	54.762
Variable construction	10.000	23.810
Sample size / data collection	12.000	28.571
Data exclusion / filtering	12.000	28.571
Design / reporting scope	25.000	59.524

Note: Counts indicate the number of articles with deviations from the pre-analysis plan (PAP) in each deviation category. Percentages are calculated relative to the total number of articles with PAP deviations ($n = 42$). Because an article may contain more than one type of deviation, counts and percentages may sum to more than the total.

A APPENDIX A

A.1 Authors' Contribution

Preparation of tables, figures, and manuscript. Abel Brodeur, Ghina Abdul Baki, Luna Fazio, Lenka Fiala, Derek Mikola.

Revisions and feedback on the manuscript. Abel Brodeur, Ghina Abdul Baki, Luna Fazio, Lenka Fiala, Derek Mikola, Mario Gollwitzer, Moin Syed, Jessica L. Arend, Thom Baguley, René Bekkers, Karlo Doroc, Joris Frese, B. Göktürk Gök, Amélie Gourdon-Kanhukamwe, Pablo Rodrigo Grassi, Lena Hahn, Mickaël Jury, Ruben Laukenmann, Charly Marie, David Nichols, Lukas Röseler, Joel Snyder, Benedikt Stöckle, Peter Szecsi, Rémi Thériault, Johannes Ullrich, David F. Urschler, David Valenta, Elisabeth Julie Vargo, Ke Wang, Daniel Woods, Mirela Zaneva.

Conception or design of the work. Abel Brodeur, Ghina Abdul Baki, Luna Fazio, Lenka Fiala, Derek Mikola, Juan Pablo Aparicio, Mario Gollwitzer, Moin Syed.

Data preparation. Abel Brodeur, Ghina Abdul Baki, Luna Fazio, Lenka Fiala, Derek Mikola, Juan Pablo Aparicio.

Analysis, cleaning and interpretation of data. Ghina Abdul Baki, Luna Fazio.

Author of reproduction report. All authors except Abel Brodeur, Ghina Abdul Baki, Lenka Fiala, Derek Mikola, Juan Pablo Aparicio, Mario Gollwitzer, Moin Syed

A.2 Open Practices Statement

All reproduction reports, data, and code associated with this project are publicly available through the Institute for Replication repository. The reproduction reports are provided in Section B.0.1, and the data and code are available at doi.org/10.5281/zenodo.22713267.

A.3 Guidelines for Choosing a Study

For the replication games, participants are assigned to a small team of about 3–5 researchers. Participants are then offered a list of articles about three weeks before the games. They are asked to choose a paper as a team. They are provided the following guidelines for choosing a study:

> Please read the Readme files to check for (i) (too) large data set/running time, (ii) software being used, (iii) completeness of the raw data.
>
> The choice of which paper to reproduce is very important. Avoid choosing a study using (i) methods you are not familiar with, (ii) use super computer or very long running time, (iii) only share final data set or (iv) data set in a language none of you can read.
>
> Last, avoid choosing a paper for which you have a conflict of interest (*e.g.*, friend, coauthor).

A.4 Coauthor Survey

INSTITUTE for REPLICATION

Psych Science Coauthor Survey

You and Your Team

* 1. Do you want to be a coauthor on I4R's Psychological Science (Psych Science) meta paper?

- ◯ Yes
- ◯ No

* 2. What is your name?

* 3. Please provide a long-term email address

* 4. What is your current affiliation (work, university, institution, *etc*.)

* 5. What is your field? (e.g., economics, psychology, sociology, etc)

* 6. How many years have you been programming?

* 7. What is your level of expertise as a programmer?

- ◯ Novice
- ◯ Advanced beginner
- ◯ Competent
- ◯ Proficient
- ◯ Expert

* 8. Have you ever had to produce a replication package?

- ◯ No
- ◯ Yes

* 9. What is your current position?

- ◯ Undergraduate Student (BA, AB, BSc, BCS, BEng, etc.)
- ◯ Master Student (MA, MSc, MEng, etc.)
- ◯ Doctoral Student (PhD, ScD, etc.)
- ◯ Postdoctoral Fellow
- ◯ Researcher with a Ph.D.
- ◯ Assistant Professor
- ◯ Associate Professor
- ◯ Full Professor
- ◯ Other (please specify)

* 10. If applicable, which year did you complete your doctorate?

- ◯ Not Applicable
- ◯ Please specify the year in which you completed your Ph.D. (in XXXX format)

* 11. If applicable, how many citations do you have on Google Scholar or equivalent website which tallies citations?

- ◯ Not applicable
- ◯ Please specify the number of citations.

INSTITUTE for REPLICATION

Psych Science Coauthor Survey

Your Replication Report

* 12. What was the title of the paper you reproduced and/or replicated?
(If multiple papers, please complete multiple surveys).

* 13. For what reasons did you select your specific paper to reproduce and/or replicate from the list of papers provided? (Select all that apply)

- [ ] Not given a choice; only one paper was given to my team
- [ ] Beliefs about statistical power/sample size
- [ ] Ex ante beliefs that main results are robust/replicable
- [ ] Ex ante beliefs that main results are not robust/replicable
- [ ] Had previously read this paper
- [ ] Length of computational time to reproduce the results
- [ ] Size of replication package
- [ ] Methods used
- [ ] Trust of original authors
- [ ] Paper's number of citations
- [ ] Avoiding conflict of interest
- [ ] Other (please specify)

* 14. What programming language(s) did you use to make your replication report? (excluding word processors like MS Word or LaTeX).

- [ ] Stata
- [ ] R
- [ ] MATLAB
- [ ] Python
- [ ] Julia
- [ ] SPSS
- [ ] Other (please specify)

* 15. How many active **days** have *you* worked on your replication report, from the day your paper was assigned to you until the final report was published as a discussion paper?

Consider it an active day if you either completed one or multiple tasks for the report (e.g., produced tables, proofread the text, fixed original code) or spent a substantial portion of your workday working on the report even if you did not finish a task (e.g., you started re-coding data cleaning).

INSTITUTE for REPLICATION

Psych Science Coauthor Survey

Original Authors' Paper and Replication Package

* 16. What is the runtime of the original authors' code (in hours)?

By runtime we mean how many hours it takes for all the codes to run, from initial cleaning until all results are completed. Do NOT include time you spent fixing the code to make it run.

* 17. What programming language(s) were used in the original authors' replication package? (Select all that apply)

- [ ] Stata
- [ ] R
- [ ] MATLAB
- [ ] Python
- [ ] Julia
- [ ] SPSS
- [ ] Other (please specify)

* 18. Which method(s) did the original paper use? (Select all that apply)

- [ ] Laboratory Experiment
- [ ] Online Experiment
- [ ] Field Experiment
- [ ] Online Survey (non-experimental)
- [ ] Simulations
- [ ] Descriptive/Non-Causal Methods
- [ ] Other (please specify)

* 19. For which of the following reasons were you unable to conduct robustness checks? (Select all that apply)

- [ ] Not applicable: we were able to conduct robustness checks
- [ ] Lack of raw data
- [ ] Lack of analytical data
- [ ] Lack of data dictionary
- [ ] Unclear documentation (e.g. how to obtain non-public data, README, etc.)
- [ ] Unclear description of data/methods in Original Authors' article
- [ ] We did not want to run robustness checks
- [ ] Other (please specify)

* 20. Did your team improve the original authors' replication package?

- ◯ No
- ◯ Yes (please specify)

* 21. Please select the category which best represents the element of the replication package for the article you reproduced/replicated:

	Provided and Clear	Provided but Unclear	Not Provided	Not Applicable
README document	◯	◯	◯	◯
Data Dictionary	◯	◯	◯	◯
How to obtain data omitted from Replication Package	◯	◯	◯	◯
Additional documentation for primary collected surveys or experimental data	◯	◯	◯	◯
Code execution instructions	◯	◯	◯	◯
Computational requirements, including libraries, packages and running time	◯	◯	◯	◯

* 22. Was your interaction with I4R (Replication Games, or report) the first time you looked at another authors' replication package?

- ◯ Yes
- ◯ No

* 23. Which of the following best describes how the replication package aligned with your expectations:

- ◯ Aligned reasonably with expectations
- ◯ Better than expected
- ◯ Worse than expected
- ◯ Had no expectations about replication packages
- ◯ Other (please specify)

* 24. How does the quality of the replication package affect your view of the discipline as a whole?

- ◯ Does not affect view of discipline
- ◯ More optimistic view of discipline
- ◯ More pessimistic view of discipline
- ◯ Other (please specify)

* 25. Select all dimensions you think your replication report will be independently (i.e., not through the meta paper) impactful:

- ☐ Will be published
- ☐ Will be cited
- ☐ Will be read
- ☐ Will circulate social media
- ☐ Will be downloaded
- ☐ Will contribute to changing practice within the field
- ☐ No impact

INSTITUTE for REPLICATION

Psych Science Coauthor Survey

Preregistration of the Original Study

The following questions capture whether the paper being reproduced was preregistered (or included a pre-analysis plan).

* 26. How many studies does the paper you reproduced have (including the pilot study)?

* 27. Was the original paper preregistered?

- ◯ Yes, all studies
- ◯ Yes, only one study/subset of studies
- ◯ No

* 28. Where was the preregistration hosted?

- ◯ OSF
- ◯ ClinicalTrials.gov
- ◯ Other (please specify)

* 29. Which elements were specified in the pre-analysis plan? (Select all that apply)

- [ ] Hypotheses
- [ ] Sample size / power calculation
- [ ] Inclusion / exclusion criteria
- [ ] Variable construction
- [ ] Primary outcome variables
- [ ] Primary predictor / treatment variable
- [ ] Other control variables
- [ ] Statistical model / estimation strategy
- [ ] Multiple hypothesis correction
- [ ] Robustness checks
- [ ] Subgroup / heterogeneity analyses
- [ ] Data cleaning procedures
- [ ] Other (please specify)

* 30. Based on your assessment, did the original paper deviate from its pre-analysis plan?

- ◯ No noticeable deviation
- ◯ Minor deviations
- ◯ Major deviations
- ◯ Not applicable (no preregistration)

* 31. If deviations occurred, were they clearly disclosed in the paper?

- ◯ Yes
- ◯ Partially
- ◯ No
- ◯ No deviations occurred

INSTITUTE for REPLICATION

Psych Science Coauthor Survey

Your Preregistration Experience and Beliefs

* 32. Approximately how many studies have you personally preregistered?

* 33. In your opinion, how much does preregistration improve the credibility of research findings? 0 = No improvement; 100 = Strongly improves credibility

0 50 100

* 34. How likely do you believe a preregistered study is to replicate (i.e., using new data) compared to a non-preregistered study?

- Likely
- Somewhat likely
- Neither likely nor unlikely
- Somewhat unlikely
- Unlikely

* 35. Did the presence of a pre-analysis plan make reproducing the results of this paper easier?

- Much easier
- Slightly easier
- No difference
- Slightly harder
- Much harder
- Not applicable (no pre-analysis plan)

B APPENDIX B

B.0.1 Report

Title Original Study: A Practical Significance Bias in Laypeople's Evaluation of Scientific Findings
doi: https://journals.sagepub.com/doi/10.1177/09567976241231506
Link to Full Report and Author Response: https://osf.io/s4et7/

B.0.2 Report

Title Original Study: A Universal Cognitive Bias in Word Order: Evidence From Speakers Whose Language Goes Against It
doi: https://journals.sagepub.com/doi/10.1177/09567976231222836
Link to Full Report and Author Response: https://osf.io/96nec/

B.0.3 Report

Title Original Study: Attitudes and Laws About Abortion Are Linked to Extrinsic Mortality Risk: A Life-History Perspective on Variability in Reproductive Rights
doi: https://journals.sagepub.com/doi/10.1177/09567976231217410
Link to Full Report and Author Response: https://osf.io/bz8pf/

B.0.4 Report

Title Original Study: Biases in Improvement Decisions: People Focus on the Relative Reduction in Bad Outcomes
doi: https://journals.sagepub.com/doi/10.1177/09567976241232891
Link to Full Report and Author Response: https://osf.io/4a2ne/

B.0.5 Report

Title Original Study: Bridging the Gap Between Self-Report and Behavioral Laboratory Measures: A Real-Time Driving Task With Inverse Reinforcement Learning
doi: https://journals.sagepub.com/doi/10.1177/09567976241228503
Link to Full Report and Author Response: https://osf.io/pua9m/

B.0.6 Report

Title Original Study: Can Invalid Information Be Ignored When It Is Detected?
doi: https://journals.sagepub.com/doi/10.1177/09567976241231571
Link to Full Report and Author Response: https://osf.io/us2g4/

B.0.7 Report

Title Original Study: Can One Donation a Day Keep Depression Away? Three Randomized Controlled Trials of an Online Micro-Charitable Giving Intervention

doi: https://journals.sagepub.com/doi/full/10.1177/09567976251315679

Link to Full Report and Author Response: https://osf.io/hxadk/

B.0.8 Report

Title Original Study: Caricaturing Shapes in Visual Memory

doi: https://journals.sagepub.com/doi/10.1177/09567976231225091

Link to Full Report and Author Response: https://osf.io/rn8t7/

B.0.9 Report

Title Original Study: Children Sustain Cooperation in a Threshold Public-Goods Game Even When Seeing Others' Outcomes (Team 1)

doi: https://journals.sagepub.com/doi/10.1177/09567976241267854

Link to Full Report and Author Response: https://osf.io/mgy45/

B.0.10 Report

Title Original Study: Children Sustain Cooperation in a Threshold Public-Goods Game Even When Seeing Others' Outcomes (Team 2)

doi: https://journals.sagepub.com/doi/10.1177/09567976241267854

Link to Full Report and Author Response: https://osf.io/mgy45/

B.0.11 Report

Title Original Study: Cutting Through the Noise: Auditory Scenes and Their Effects on Visual Object Processing

doi: https://journals.sagepub.com/doi/10.1177/09567976241237737

Link to Full Report and Author Response: https://osf.io/n5dz2/

B.0.12 Report

Title Original Study: Deconfounding Sex and Sex of Partner in Mate-Preference Research

doi: https://journals.sagepub.com/doi/10.1177/09567976251319038

Link to Full Report and Author Response: https://osf.io/j65xf/

B.0.13 Report

Title Original Study: Devaluation by Omission: Limited Identity Options Elicit Anger and Increase Identification

doi: https://journals.sagepub.com/doi/10.1177/09567976231223416

Link to Full Report and Author Response: https://osf.io/w8gd7/

B.0.14 Report

Title Original Study: Dissociable Codes in Motor Working Memory
doi: https://journals.sagepub.com/doi/10.1177/09567976231221756
Link to Full Report and Author Response: https://osf.io/qe3fx/

B.0.15 Report

Title Original Study: Directing Attention Shapes Learning in Adults but Not Children
doi: https://journals.sagepub.com/doi/10.1177/09567976241263347
Link to Full Report and Author Response: https://osf.io/gjm3c/

B.0.16 Report

Title Original Study: Disagreement Gets Mistaken for Bad Listening
doi: https://journals.sagepub.com/doi/10.1177/09567976241239935
Link to Full Report and Author Response: https://osf.io/k6ntc/

B.0.17 Report

Title Original Study: Distinct Inhibitory-Control Processes Underlie Children's Judgments of Fairness
doi: https://journals.sagepub.com/doi/10.1177/09567976231217420
Link to Full Report and Author Response: https://osf.io/pzhgw/

B.0.18 Report

Title Original Study: Do We Become More Lonely With Age? A Coordinated Data Analysis of Nine Longitudinal Studies
doi: https://journals.sagepub.com/doi/10.1177/09567976241242037
Link to Full Report and Author Response: https://osf.io/ejsb9/

B.0.19 Report

Title Original Study: Does Physiological Arousal Increase Social Transmission of Information? Two Replications of Berger (2011)
doi: https://journals.sagepub.com/doi/10.1177/09567976241257255
Link to Full Report and Author Response: https://osf.io/grvnq/

B.0.20 Report

Title Original Study: Does Valuing Happiness Lead to Well-Being?
doi: https://journals.sagepub.com/doi/10.1177/09567976241263784
Link to Full Report and Author Response: https://osf.io/9yve3/

B.0.21 Report

Title Original Study: Effects of Voice Pitch on Social Perceptions Vary With Relational Mobility and Homicide Rate

doi: https://journals.sagepub.com/doi/10.1177/09567976231222288

Link to Full Report and Author Response: https://osf.io/ts2mp/

B.0.22 Report

Title Original Study: Exploring the Semantic-Inconsistency Effect in Scenes Using a Continuous Measure of Linguistic-Semantic Similarity

doi: https://journals.sagepub.com/doi/10.1177/09567976241238217

Link to Full Report and Author Response: https://osf.io/mhrf7/

B.0.23 Report

Title Original Study: Gaze Behavior Reveals Expectations of Potential Scene Changes

doi: https://journals.sagepub.com/doi/10.1177/09567976241279198

Link to Full Report and Author Response: https://osf.io/6smqd/

B.0.24 Report

Title Original Study: Gender Differences in Climbing up the Ladder: Why Experience Closes the Ambition Gender Gap

doi: https://journals.sagepub.com/doi/10.1177/09567976241266778

Link to Full Report and Author Response: https://osf.io/rcngs/

B.0.25 Report

Title Original Study: Gender-Ambiguous Voices and Social Disfluency

doi: https://journals.sagepub.com/doi/10.1177/09567976241238222

Link to Full Report and Author Response: https://osf.io/u2rx8/

B.0.26 Report

Title Original Study: Generating New Musical Preferences From Multilevel Mapping of Predictions to Reward

doi: https://journals.sagepub.com/doi/10.1177/09567976231214185

Link to Full Report and Author Response: https://osf.io/hkuns/

B.0.27 Report

Title Original Study: Identity Concealment May Discourage Health-Seeking Behaviors: Evidence From Sexual-Minority Men During the 2022 Global Mpox Outbreak

doi: https://journals.sagepub.com/doi/10.1177/09567976231217416

Link to Full Report and Author Response: https://osf.io/9jzvk/

B.0.28 Report

Title Original Study: Investigating Inattentional Blindness Through the Lens of Fear Chemosignals
doi: https://journals.sagepub.com/doi/full/10.1177/09567976231213572
Link to Full Report and Author Response: https://osf.io/hbpcs/

B.0.29 Report

Title Original Study: Is Mansplaining Gendered? The Effects of Unsolicited, Generic, and Prescriptive Advice on U.S. Women
doi: https://journals.sagepub.com/doi/full/10.1177/09567976241268630
Link to Full Report and Author Response: https://osf.io/2pqdz/

B.0.30 Report

Title Original Study: Learning From Aggregated Opinion
doi: https://journals.sagepub.com/doi/10.1177/09567976241251741
Link to Full Report and Author Response: https://osf.io/7cd2s/

B.0.31 Report

Title Original Study: Manipulating Prior Beliefs Causally Induces Under- and Overconfidence
doi: https://journals.sagepub.com/doi/full/10.1177/09567976241231572
Link to Full Report and Author Response: https://osf.io/jnzes/

B.0.32 Report

Title Original Study: Mindful-Gratitude Practice Reduces Prejudice at High Levels of Collective Narcissism
doi: https://journals.sagepub.com/doi/10.1177/09567976231220902
Link to Full Report and Author Response: https://osf.io/8czs5/

B.0.33 Report

Title Original Study: Misplaced Divides? Discussing Political Disagreement With Strangers Can Be Unexpectedly Positive
doi: https://journals.sagepub.com/doi/10.1177/09567976241230005
Link to Full Report and Author Response: https://osf.io/rkdqv/

B.0.34 Report

Title Original Study: Narrative Identity, Traits, and Trajectories of Depression and Well-Being: A 9-Year Longitudinal Study

doi: https://journals.sagepub.com/doi/full/10.1177/09567976241296512

Link to Full Report and Author Response: https://osf.io/xqnhw/

B.0.35 Report

Title Original Study: Not All Powerful People Are Created Equal: An Examination of Gender and Pathways to Social Hierarchy Through the Lens of Social Cognition

doi: https://journals.sagepub.com/doi/full/10.1177/09567976241260251

Link to Full Report and Author Response: https://osf.io/g82fy/

B.0.36 Report

Title Original Study: On the Efficacy of Accuracy Prompts Across Partisan Lines: An Adversarial Collaboration

doi: https://journals.sagepub.com/doi/10.1177/09567976241232905

Link to Full Report and Author Response: https://osf.io/usbjn/

B.0.37 Report

Title Original Study: On the Robustness and Provenance of the Gambler's Fallacy

doi: https://journals.sagepub.com/doi/10.1177/09567976251344570

Link to Full Report and Author Response: https://osf.io/5c9aj/

B.0.38 Report

Title Original Study: Participating in a Digital-History Project Mobilizes People for Symbolic Justice and Better Intergroup Relations Today

doi: https://journals.sagepub.com/doi/10.1177/09567976251331040

Link to Full Report and Author Response: https://osf.io/3fy5c/

B.0.39 Report

Title Original Study: People Endorse Harsher Policies in Principle Than in Practice: Asymmetric Beliefs About Which Errors to Prevent Versus Fix

doi: https://journals.sagepub.com/doi/10.1177/09567976241228504

Link to Full Report and Author Response: https://osf.io/wg2sv/

B.0.40 Report

Title Original Study: People Place Larger Bets When Risky Choices Provide a Postbet Option to Cash Out

doi: https://journals.sagepub.com/doi/10.1177/09567976241266516

Link to Full Report and Author Response: https://osf.io/fpaxs/

B.0.41 Report

Title Original Study: Perceiving Topological Relations

doi: https://journals.sagepub.com/doi/10.1177/09567976241309615

Link to Full Report and Author Response: https://osf.io/xg3h9/

B.0.42 Report

Title Original Study: Personal Misconduct Elicits Harsher Professional Consequences for Artists (vs. Scientists): A Moral-Decoupling Process

doi: https://journals.sagepub.com/doi/10.1177/09567976231214739

Link to Full Report and Author Response: https://osf.io/jybu2/

B.0.43 Report

Title Original Study: Preregistered Replication and Extension of "Moral Hypocrisy: Social Groups and the Flexibility of Virtue"

doi: https://journals.sagepub.com/doi/10.1177/09567976241246552

Link to Full Report and Author Response: https://osf.io/mz8gt/

B.0.44 Report

Title Original Study: Proenvironmental Behavior Increases Subjective Well-Being: Evidence From an Experience-Sampling Study and a Randomized Experiment

doi: https://journals.sagepub.com/doi/10.1177/09567976241251766

Link to Full Report and Author Response: https://osf.io/9c8p6/

B.0.45 Report

Title Original Study: Racial Minorities Face Discrimination From Across the Political Spectrum When Seeking to Form Ties on Social Media: Evidence From a Field Experiment

doi: https://journals.sagepub.com/doi/full/10.1177/09567976241274738

Link to Full Report and Author Response: https://osf.io/d5y3e/

B.0.46 Report

Title Original Study: Racial Prejudice Affects Representations of Facial Trustworthiness
doi: https://journals.sagepub.com/doi/10.1177/09567976231225094
Link to Full Report and Author Response: https://osf.io/529en/

B.0.47 Report

Title Original Study: Reducing facial stereotype bias in consequential social judgments: Intervention success with White male faces
doi: https://journals.sagepub.com/doi/10.1177/09567976231215238
Link to Full Report and Author Response: https://osf.io/qt7kr/

B.0.48 Report

Title Original Study: Rethinking the Role of Teams and Training in Geopolitical Forecasting: The Effect of Uncontrolled Method Variance on Statistical Conclusions
doi: https://journals.sagepub.com/doi/10.1177/09567976241266481
Link to Full Report and Author Response: https://osf.io/vjka5/

B.0.49 Report

Title Original Study: Risky-Choice Framing Effects Result Partly From Mismatched Option Descriptions in Gains and Losses
doi: https://journals.sagepub.com/doi/10.1177/09567976241249183
Link to Full Report and Author Response: https://osf.io/t68dc/

B.0.50 Report

Title Original Study: "Stop the Count!"—How Reporting Partial Election Results Fuels Beliefs in Election Fraud
doi: https://journals.sagepub.com/doi/10.1177/09567976251355594
Link to Full Report and Author Response: https://osf.io/82c6v/

B.0.51 Report

Title Original Study: Synchrony Influences Estimates of Cooperation in a Public-Goods Game
doi: https://journals.sagepub.com/doi/full/10.1177/09567976231223410
Link to Full Report and Author Response: https://osf.io/8tzx5/

B.0.52 Report

Title Original Study: The Affect Misattribution Procedure Revisited: An Informational Account
doi: https://journals.sagepub.com/doi/10.1177/09567976241287735
Link to Full Report and Author Response: https://osf.io/gh7cv/

B.0.53 Report

Title Original Study: The Double-Edged Sword of Social Sharing: Social Sharing Predicts Increased Emotion Differentiation When Rumination Is Low but Decreased Emotion Differentiation When Rumination Is High

doi: https://journals.sagepub.com/doi/10.1177/09567976241266513

Link to Full Report and Author Response: https://osf.io/jtzxp/

B.0.54 Report

Title Original Study: The Effects of Group Affiliation Versus Individuating Information on Direct and Indirect Measures of the Evaluation of Novel Individual Group Members

doi: https://journals.sagepub.com/doi/10.1177/09567976251315689

Link to Full Report and Author Response: https://osf.io/6mzpv/

B.0.55 Report

Title Original Study: The Extended Mind in Young Children: Cost-Dependent Trade-Off Between External and Internal Memory

doi: https://journals.sagepub.com/doi/10.1177/09567976241306424

Link to Full Report and Author Response: https://osf.io/3b8z4/

B.0.56 Report

Title Original Study: The Formation of Negative Attitudes Toward Novel Groups

doi: https://journals.sagepub.com/doi/10.1177/09567976241239932

Link to Full Report and Author Response: https://osf.io/av3mz/

B.0.57 Report

Title Original Study: The Language of (Non)Replicable Social Science

doi: https://journals.sagepub.com/doi/10.1177/09567976241254037

Link to Full Report and Author Response: https://osf.io/kpm3d/

B.0.58 Report

Title Original Study: The Interactive Effect of Incentive Salience and Prosocial Motivation on Prosocial Behavior (Team 1)

doi: https://journals.sagepub.com/doi/abs/10.1177/09567976241234560

Link to Full Report and Author Response: https://osf.io/q9vw7/

B.0.59 Report

Title Original Study: The Interactive Effect of Incentive Salience and Prosocial Motivation on Prosocial Behavior (Team 2)

doi: https://journals.sagepub.com/doi/abs/10.1177/09567976241234560

Link to Full Report and Author Response: https://osf.io/q9vw7/

B.0.60 Report

Title Original Study: The Motion-Silencing Illusion Depends on Object-Centered Representation

doi: https://journals.sagepub.com/doi/10.1177/09567976241235104

Link to Full Report and Author Response: https://osf.io/s59j2/

B.0.61 Report

Title Original Study: Verbal Aggressions Against Major League Baseball Umpires Affect Their Decision Making

doi: https://journals.sagepub.com/doi/10.1177/09567976241227411

Link to Full Report and Author Response: https://osf.io/r6p98/

B.0.62 Report

Title Original Study: Visual Preference for Socially Relevant Spatial Relations in Humans and Monkeys

doi: https://journals.sagepub.com/doi/10.1177/09567976241242995

Link to Full Report and Author Response: https://osf.io/auw49/

B.0.63 Report

Title Original Study: When and Why People Conceal Infectious Disease

doi: https://journals.sagepub.com/doi/10.1177/09567976231221990

Link to Full Report and Author Response: https://osf.io/nh568/

B.0.64 Report

Title Original Study: White by Another Name? Can Anti-Christian Bias Claims Serve as a Racial Dog Whistle?

doi: https://journals.sagepub.com/doi/full/10.1177/09567976241236162

Link to Full Report and Author Response: https://osf.io/natcr/

B.0.65 Report

Title Original Study: Why Do Children Think Words Are Mutually Exclusive? (Team 1)

doi: https://journals.sagepub.com/doi/10.1177/09567976241287732

Link to Full Report and Author Response: https://osf.io/auyq5/

B.0.66 Report

Title Original Study: Why Do Children Think Words Are Mutually Exclusive? (Team 2)
doi: https://journals.sagepub.com/doi/10.1177/09567976241287732
Link to Full Report and Author Response: https://osf.io/auyq5/

B.0.67 Report

Title Original Study: Why Twitter Sometimes Rewards What Most People Disapprove of: The Case of Cross-Party Political Relations
doi: https://journals.sagepub.com/doi/10.1177/09567976241258149
Link to Full Report and Author Response: https://osf.io/ym9dn/